WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending 17 October 2026 | 11 October 2026 to 17 October 2026 | 123 records pooled from 7 daily issues

123
RECORDS POOLED
(123 UNIQUE DOIs)

39%
MEASUREMENT SIDE
(48 RECORDS)

10
SUBTOPICS
ACTIVE

28
POOLED EFFECT
ESTIMATES (28 WITH A
CI)

26
ABSTRACT-FREE
(21% OF WEEK)

The week in one paragraph

Volume recovered 41 % to 123 records, the abstract gap narrowed to 21 %, and the health side came back — but the week's most consequential records are about the reference, not the sensor: the gravimetric filter, the lidar retrieval and the drone platform each carry a bias that propagates into every calibration built on them.

Scale. 123 records over seven issues on daily totals of **1**, 24, 16, 24, **25**, 18 and 15. The single-record Sunday (11 October) is a weekend-deposit trough, not a harvest failure. Against recent rollups: 123 after 87, 125, 162, 148, 126 and 150. **All ten subtopics are populated** for the fourth week running.
The measurement side held at 39 %. *Exposure assessment & modelling* 23 → **26** and *Sensing* 15 → **22** both grew, but the health side grew faster, so the share reads 40, 44, 33, 42, 44, **39 %** across six weeks.
The health side recovered from last week's collapse. *Cardiovascular & metabolic* 1 → **7**, *Reproductive & developmental* 1 → **6**, *Respiratory & allergic* 3 → **8**. Part of this is the three held 9 October records (CHD proteomics, green-space ageing, alveolar organoids) entering the 12 October issue.
Estimates: 28, every one with an interval. One per 4.4 records (series 2.7, 3.1, 5.2, 4.2, 5.4, **4.4**); **28 of 28 with a CI** for the fourth complete week. 23 sit on ratio scales and 5 on difference scales (three PAFs, two life-expectancy coefficients).
Abstract gap: 26 of 123 (21 %), down from the 31 % series high, mainly because the Crossref title-match and Europe PMC core-lookup back-fill introduced on 17 October now recovers same-day DOIs and abstracts that previously shipped as metadata-only.
Tier mix shifted down: **15 A, 43 B, 65 C** — tier A at 12 %, from last week's series high of 21 %. More volume, more marginal records; the A-list is shorter than the week deserves.

Subtopic trend and inflection

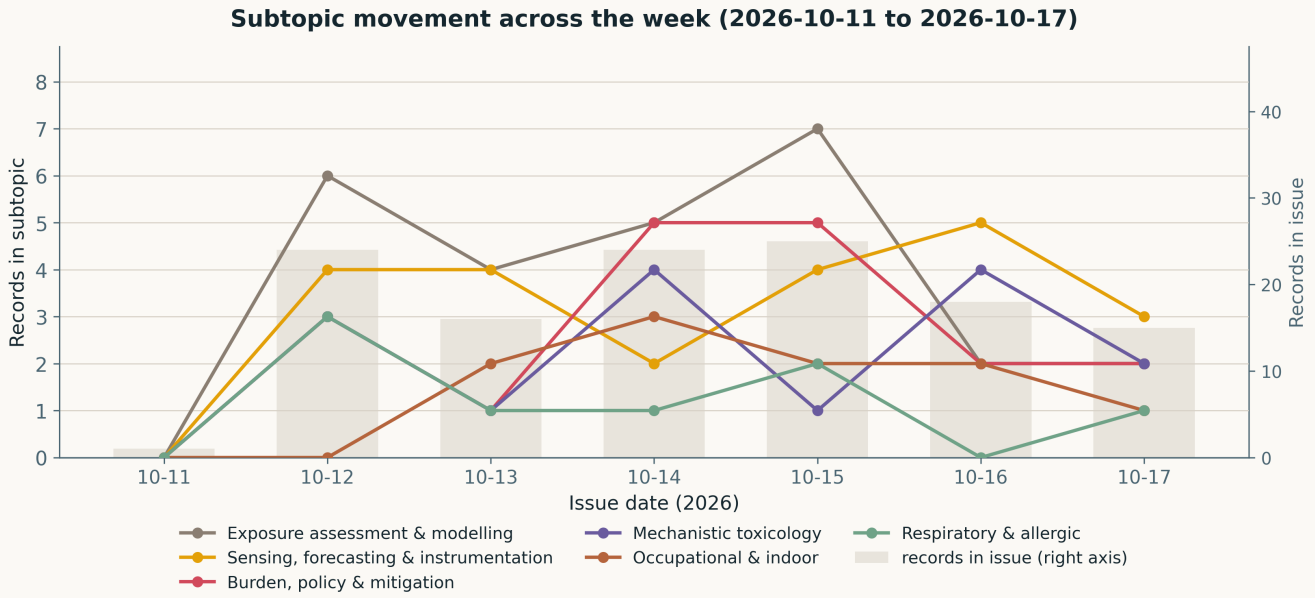


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals 1, 24, 16, 24, 25, 18 and 15. The 11 October issue was built on the 12 October run, 12–14 October on the 16 October backfill, 15 and 16 October on their next-day runs, and 17 October together with this rollup on the 19 October run — the Saturday 23:00 run did not fire.

Expanding. *Sensing* 15 → **22** is the largest absolute gain, and its content changed: retrieval and reference-method work (lunar AERONET AOD, aerosol-type-gated lidar, UAV rotor-wash bias, gravimetric filter water) displaced the routine LCS calibration papers that have dominated the band since August. *Mechanistic toxicology* 10 → **15** and *Burden, policy & mitigation* 14 → **18** also grew.

Contracting. *Neuro / mental health* 6 → **4**, though two of the four are tier A (smoke-carbonaceous ADRD in Medicare; prenatal UFP and postpartum depression in Shanghai). *Occupational & indoor* flat at 10.

The inflection worth naming. Wildfire smoke is separating from ambient $PM_{2.5}$ as an exposure in its own right — with its own composition, its own lag structure and its own endpoints. In one week: incident ADRD in 39.7 million Medicare beneficiaries at HR **1.029** (1.024–1.035) per 1 $\mu g\,m^{-3}$ smoke $PM_{2.5}$ but **1.090** (1.077–1.103) per 1 $\mu g\,m^{-3}$ of smoke EC+OC; head-and-neck cancer incidence RR **1.41** (1.24–1.60) for any annual smoke exposure over 0–10 years in SEER counties; and a deconvolved US mortality response that peaks **three years** after a smoke pulse and sums to 1.7 million deaths 2011–2023. Carbonaceous fraction, multi-year lags and non-respiratory endpoints are three things a mass-only, daily-resolved surveillance record cannot carry. The sensing side has a matching weakness: smoke PM is concentrated near 0.1–0.3 μm, where optical LCS under-count, and the US smoke corrections in use are mass-only. **The test for next week**: a smoke-specific estimate outside North America, or a validated carbonaceous channel (multi-wavelength absorption) in a network setting.

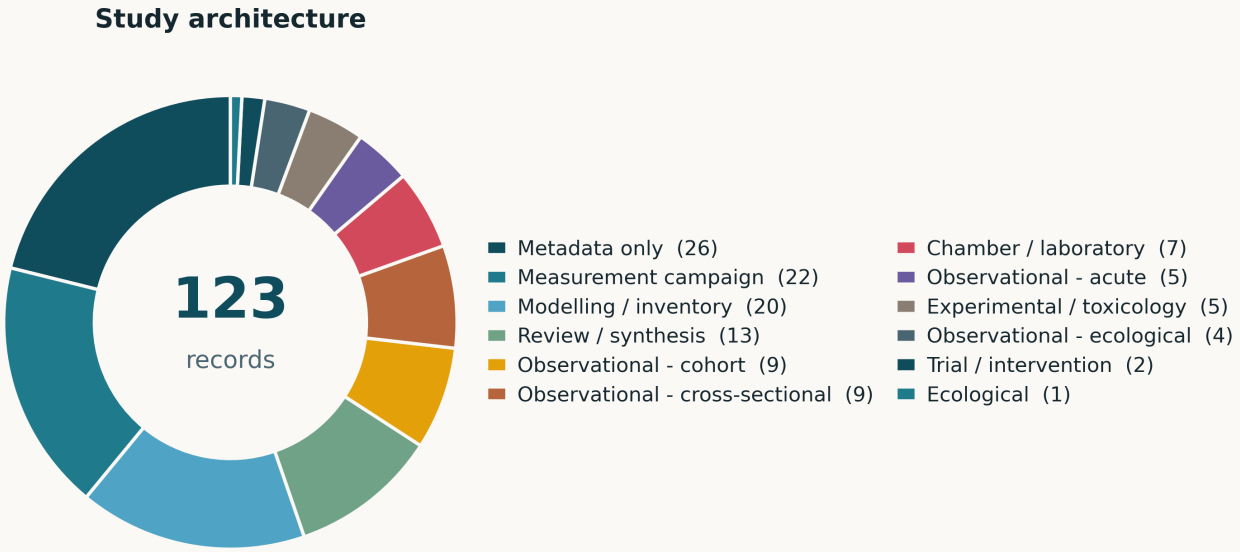


Fig. Study architecture over the pooled week. *Metadata only* 26 and *Measurement campaign* 22 lead, then *Modelling / inventory* 20 and *Review / synthesis* 13. Primary observational epidemiology (cohort 9, acute 5, cross-sectional 9, ecological 5) totals **28**, up from 16. **Two interventional records entered** — the LPG cooking RCT (quality of life) and the HEPA crossover trial carried in the 18 October issue falls in next week's pool.

Pooled analytics

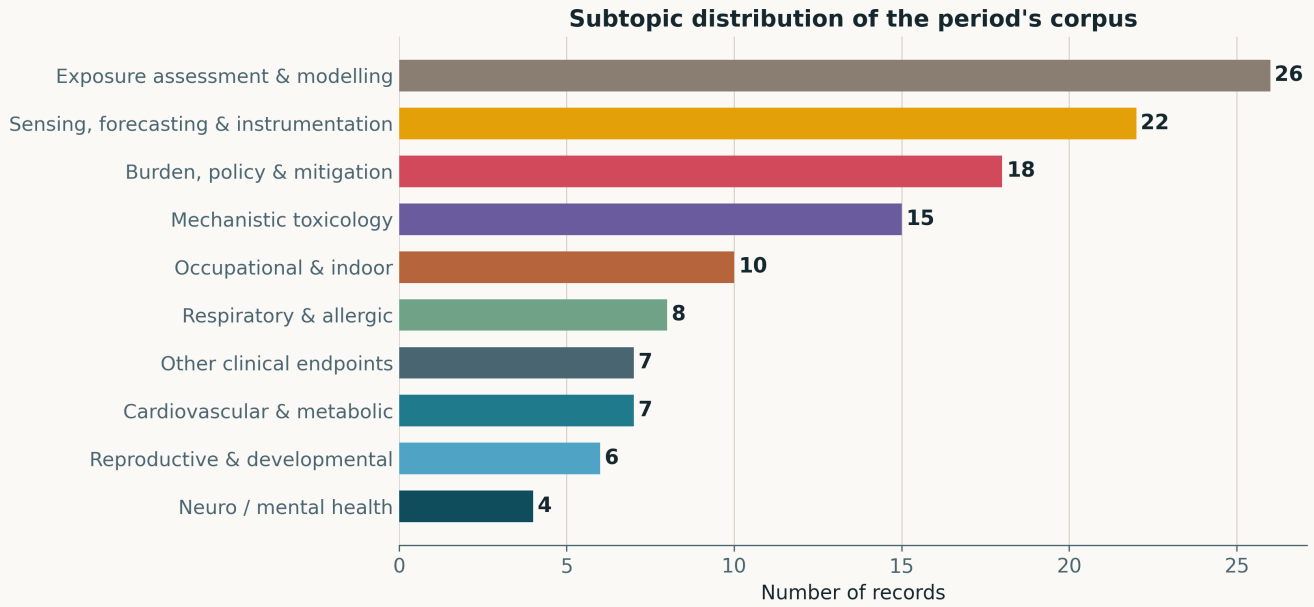


Fig. Pooled subtopic distribution, all ten bands populated: *Exposure* 26 and *Sensing* 22 put the measurement side at **48 of 123 (39 %)**. On the health side *Burden* 18 and *Mechanistic toxicology* 15 lead, then *Occupational & indoor* 10 and *Respiratory* 8; *Neuro* is the smallest band at 4.

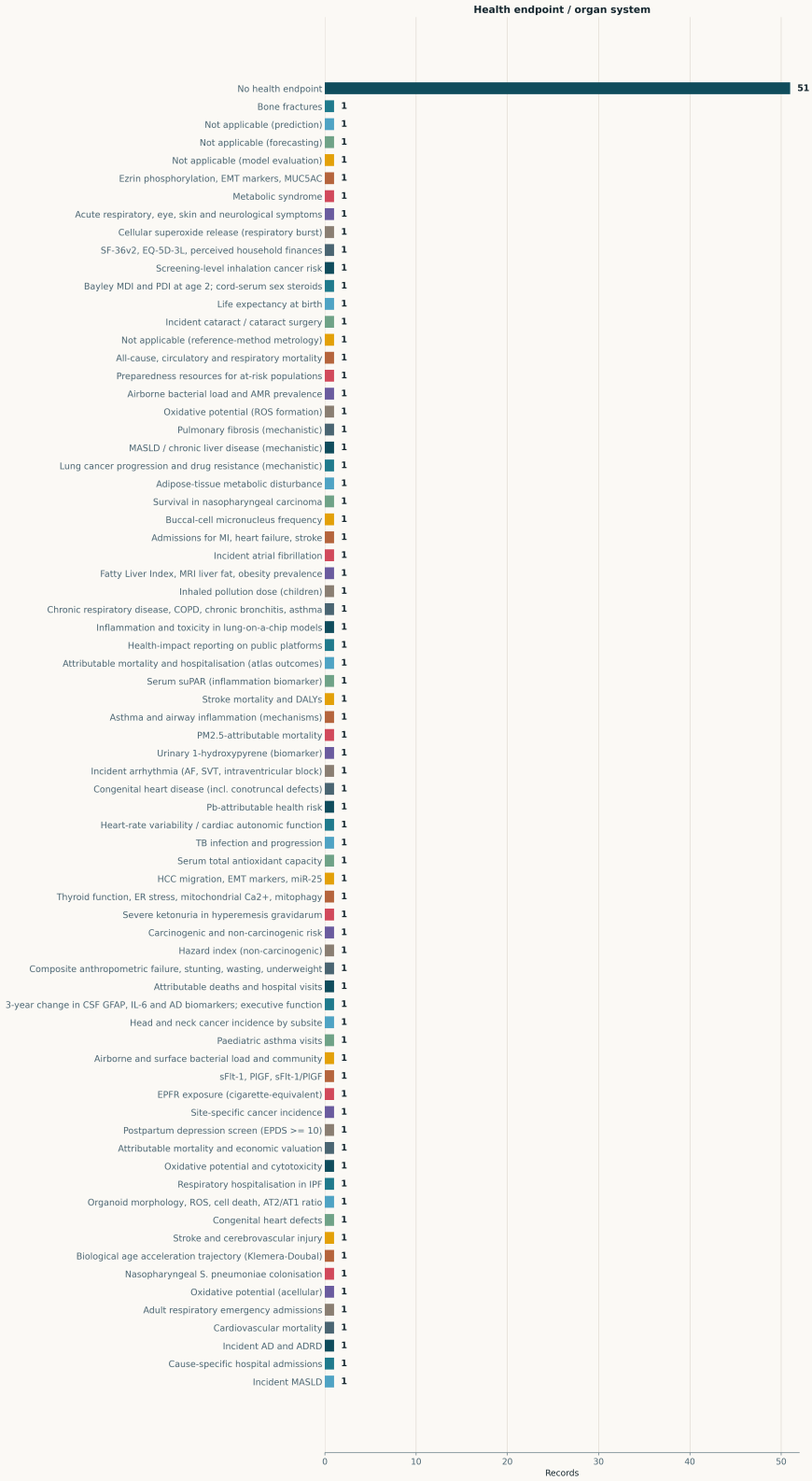


Fig. Health endpoint over the pooled week, canonicalised. "No health endpoint" pools the metadata-only records with the instrument, retrieval, emissions and modelling work. Among human outcomes mortality, admissions and neurological outcomes lead; cancer appears twice (head-and-neck, nasopharyngeal survival), both in wildfire- or biomass-burning settings.

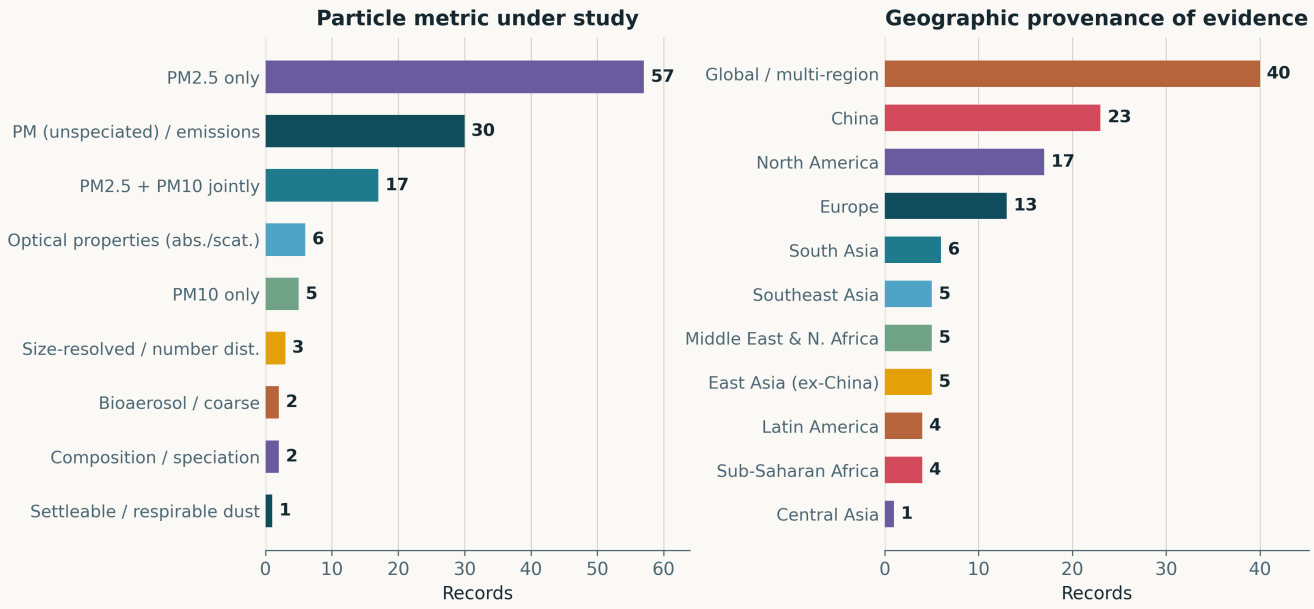


Fig. Left — particle metric. **$PM_{2.5}$ alone 57 of 123**, unspeciated PM or emissions 30, joint $PM_{2.5}$ + PM_{10} 17; optical properties 6 (lidar, lunar AOD, MISR), PM_{10} alone 5, size-resolved number 3, composition 2, bioaerosol 2, respirable dust 1. Number and composition records fell from 10 to 5 — last week's "mass is the wrong metric" signal did not persist. Right — geography. *Global / multi-region* 40 is inflated by reviews, laboratory work and 14 records whose country is not recoverable; **China 23**, North America 17, Europe 13, South Asia 6, and five each for East Asia (ex-China), MENA and Southeast Asia.

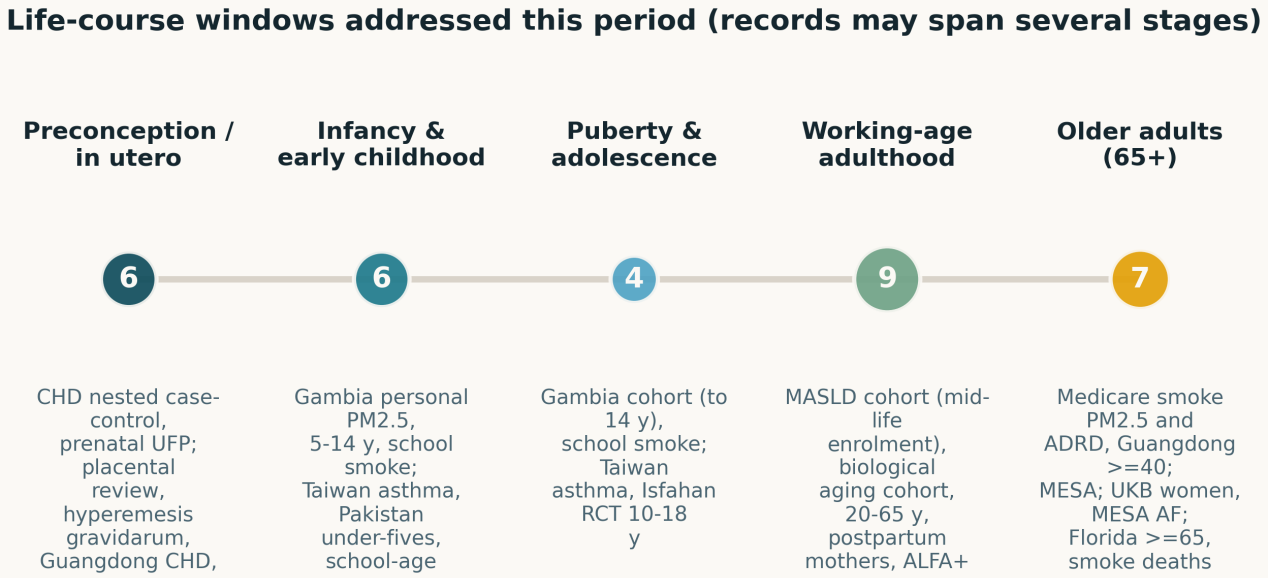


Fig. Life-course windows across the pooled week; counts non-exclusive. **Working age 9**, older adults 7, **preconception / in utero 6** (from 1: Guangdong CHD constituents, Shanghai prenatal UFP, Wuhan cohort among them), infancy & early childhood 6, **adolescence 4** — the first week since August in which no window is in single figures below 4.

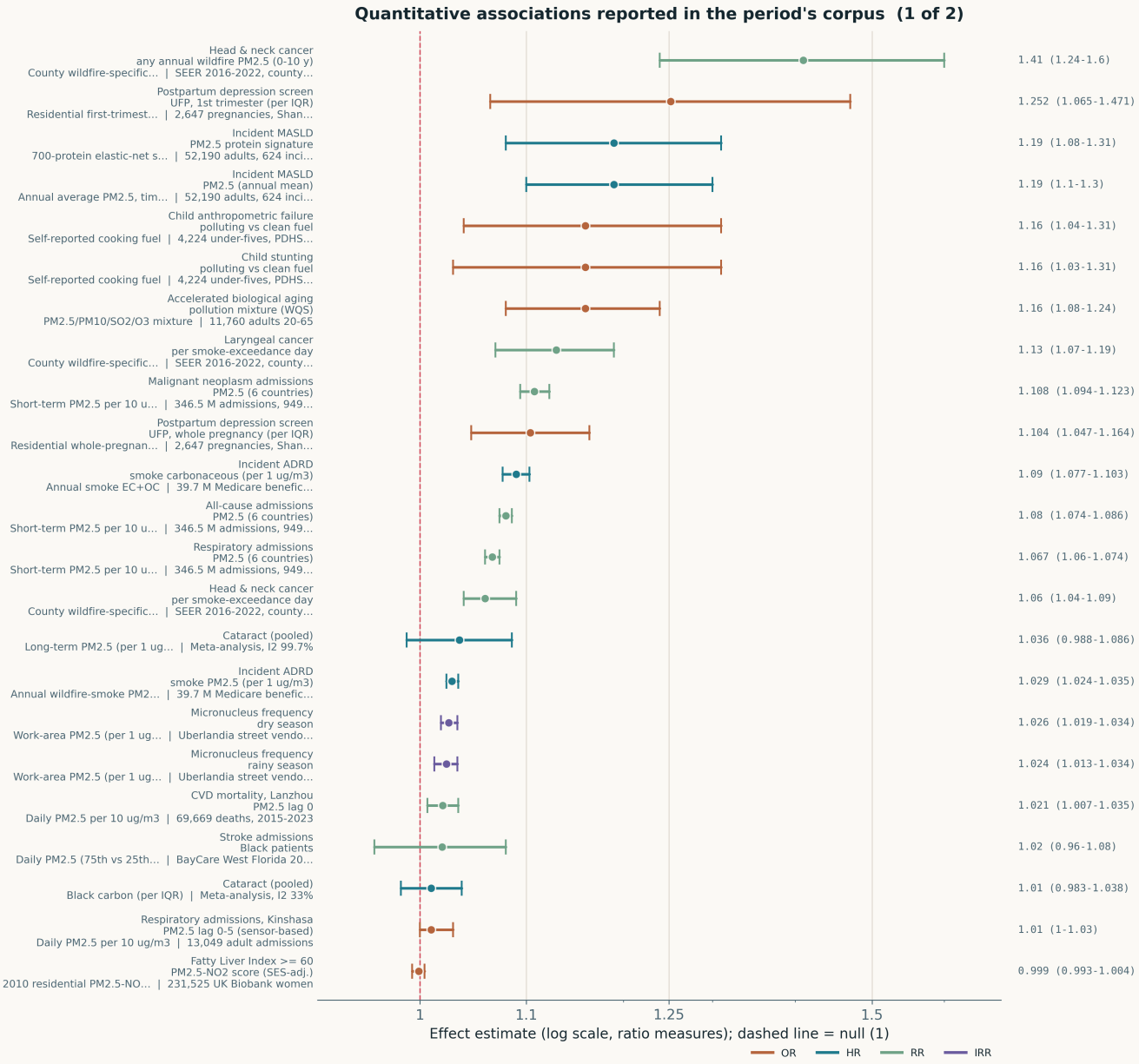


Fig. The **23** ratio-scale estimates, every one with a 95 % CI. **Not mutually comparable and not meta-analysed**: increments include per 1, per 10 μg m^{-3}, per IQR, per smoke-exceedance day, binary fuel contrasts and a mixture index, on HR, OR, RR and IRR scales, so no I^2 is reported. Readable points: the largest is head-and-neck cancer RR **1.41** (1.24–1.60); the most precise are the six-country admissions RRs from 346.5 million admissions (all-cause **1.080**, 1.074–1.086, per 10 μg m^{-3}). **Five intervals touch or cross unity**: Kinshasa respiratory admissions on sensor-based $PM_{2.5}$ (1.01, 1.00–1.03), West Florida stroke in Black patients (1.02, 0.96–1.08), the SES-adjusted fatty-liver OR (0.999), and both pooled cataract HRs. Nested families: three SEER head-and-neck RRs, two Medicare ADRD HRs, two Shanghai UFP ORs, two Uberlândia IRRs and two cataract HRs share cases within each family.

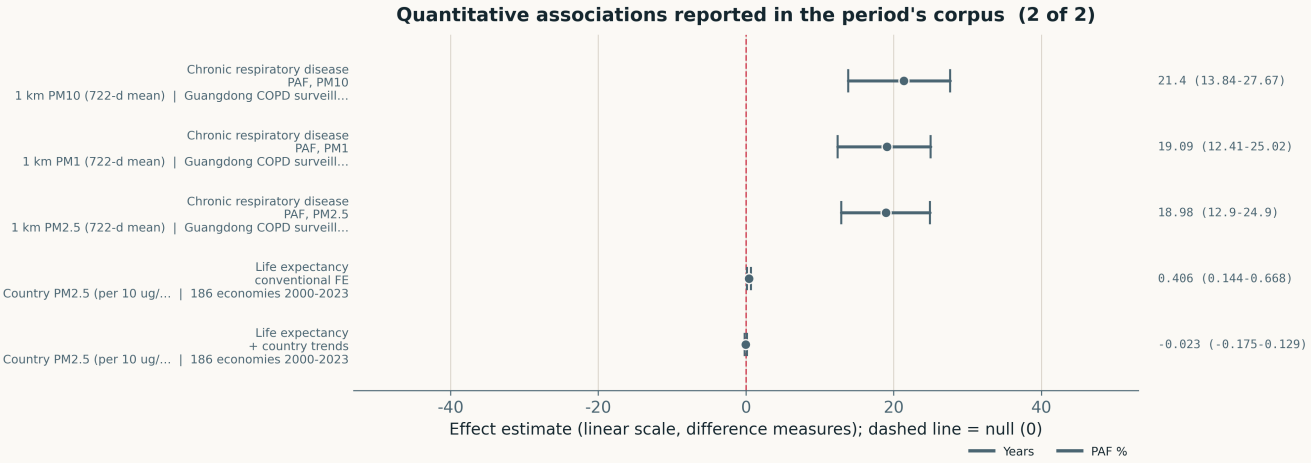


Fig. The **5** difference-scale estimates. Three population-attributable fractions (19–21 %) for chronic respiratory disease from one Guangdong sample and one 1 km satellite-ML product — one estimate drawn three times — and two life-expectancy coefficients from one 186-economy panel whose sign flips from **+0.41** to **−0.02** y per 10 μg m^{-3} once country trends are modelled.

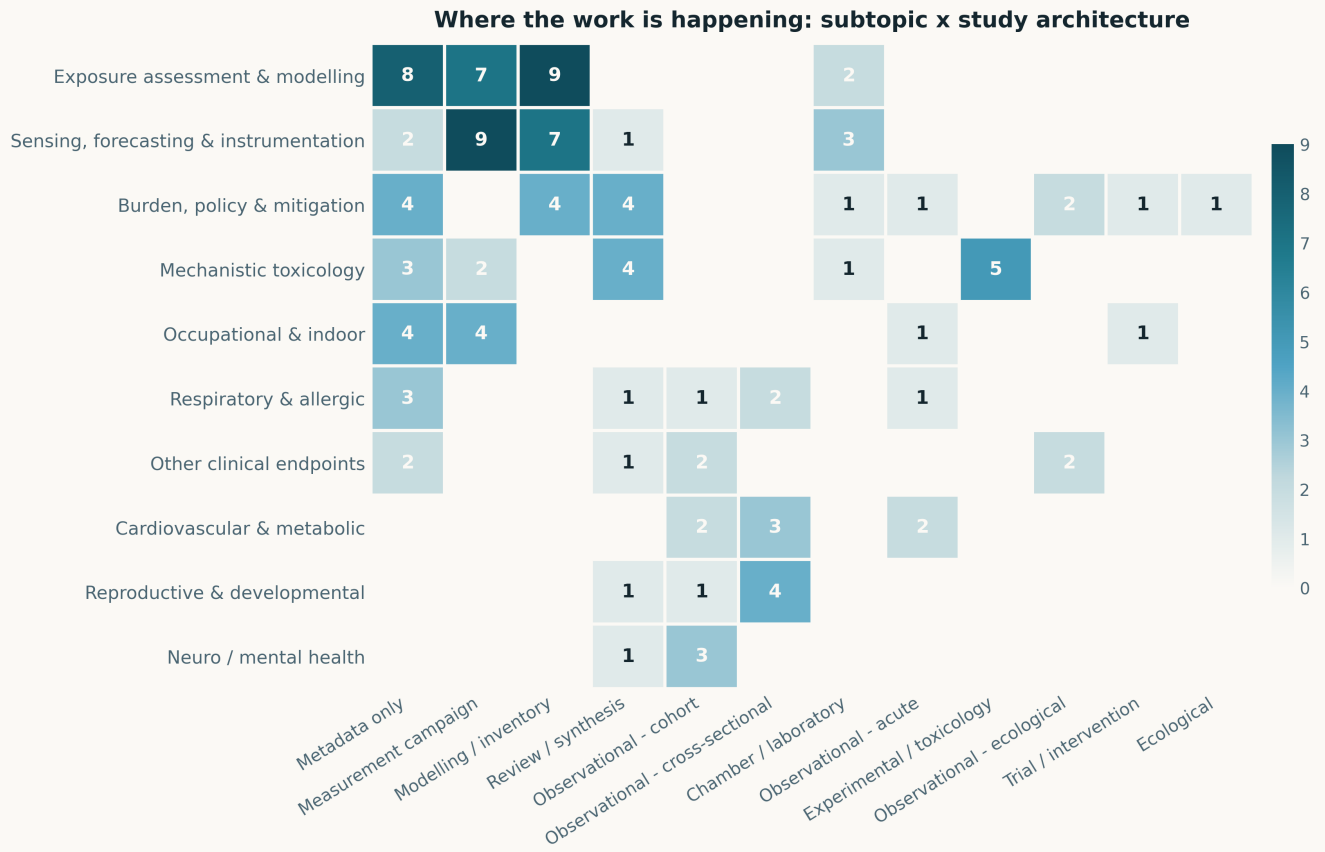


Fig. Subtopic × architecture, pooled. The densest cells remain *Exposure* × *Metadata only* and *Sensing* × *Measurement campaign*. Burden is spread across modelling and ecological designs; mechanism is split between reviews and in vitro / animal work.

Venues, trial watch, and what the pipeline did

Venues. *Environ Pollut* leads at **11** (9 %), then *Atmosphere* 7, *Atmos Chem Phys* 6, *Front Public Health* 5, and *Environ Sci Technol*, *Atmos Environ* and *Atmos Pollut Res* at 4. **Preprints supplied 10** (Research Square 4, medRxiv returning after a blank week with the smoke-mortality deconvolution).

Recurring authors. Surname frequency remains a naming artefact; **no repeat-group cluster is claimed**. The largest datasets of the week — 346.5 million admissions across six countries, 39.7 million Medicare beneficiaries, SEER 2016–2022 — each supplied one record.

Trial watch — nothing new. ClinicalTrials.gov was queried over `LastUpdatePostDate` 12–14, 16–17 and 16–19 October: **0 hits** in every window. `trials.json` holds **17 trials** over **31 logged windows**. The only interventional evidence of the week came from the literature (LPG RCT, 17 October), not the registry.

What the harvesting legs did. Crossref-by-ISSN again the highest-yield leg (36–78/day). **Europe PMC** ranged 0–123 raw/day, with the AGRICOLA retro-index again inflating two days (57 and 46 dropped); it returned 0 on the 14th. **PubMed connector** appended 3–5 records/day the harvester missed. **OpenAlex** HTTP 429 on every run — dead, not degraded. **arXiv** answered and contributed nothing in-window. **Consensus** 10 hits, **0 new** through 17 October (one new record surfaced on the 19 October sweep and is filed in the 18 October issue). **Scholar Gateway** not retried this week after two weeks of `ACCESS_DENIED`. The DOI gate caught one already-shipped record (the Washington DC 1 km CMAQ study) on 16 October.

Open gaps

- **Reference-method error is under-propagated.** Gravimetric filter water adds +3 to +18 % to PM_1/$PM_{2.5}$; unconstrained lidar extinction-to-mass conversion is 38 % worse in RMSE than a type-gated one; UAV rotor wash biases airborne readings 3–15 %. No calibration paper in the corpus carries reference uncertainty into its reported sensor RMSE.
- **Wildfire smoke needs a composition channel.** The week's smoke estimates are strongest for the carbonaceous fraction; no network record measures it.
- **The abstract gap is smaller but structural**: 26 records, concentrated in Elsevier same-day deposits on the measurement side. Scholar Gateway still needs reconnecting at the user's end.
- **Tier C at 53 %** of the week. Higher volume brought more peripheral records (country panels, reviews, metadata-only); the threshold for admission is unchanged, but the rollup is noisier.
- **Held queue empty.** The three 9 October holds entered the 12 October issue; nothing is held at period close.
- **Pipeline defects carried forward**: no first-author affiliation field; the `Cancer/Oncologic` split in `ENDPOINT_CANON`; Saturday 23:00 runs not firing (17 October built two days late); `.git` lock debris.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation — 22 records

Carried from the daily issues — full summaries (22 of 22 records in this subtopic)

Bachour WRF-Chem AOD post-correction hot desert 2026 Atmos Pollut Res

Statistical post-correction of WRF-Chem aerosol optical depth under hot-desert conditions

metadata only. No abstract at harvest. Title indicates statistical bias correction of WRF-Chem AOD in a hot-desert setting (author group based in Qatar). Recorded for coverage; no estimate inferred. Relevance: moderate - dust-dominated AOD is where both models and optical LCS fail, and corrected CTM fields are common fusion inputs.

doi: 10.1016/j.apr.2026.103233

CONIOPOL CL61 aerosol typing 2026 Atmos Meas Tech

New operational perspective to identify aerosols and hydrometeors in real-time with a pioneering algorithm (CONIOPOL) based on single-wavelength polarization lidar (CL61)

Automated classification from a polarised automatic lidar-ceilometer combining depolarisation, attenuated backscatter and cloud base to separate aerosol, cloud phase and precipitation and to type aerosol as dust, smoke/marine, continental or mixed. One year at Uccle showed strong temporal and vertical correspondence with CAMS for transported dust and smoke. Limitation: single wavelength cannot separate smoke from marine aerosol, and validation against CAMS is model-conditioned agreement rather than in-situ truth. Relevance: high - ceilometers are already networked across Europe; vertical typing tells a surface sensor network when its column assumption is broken.

doi: 10.5194/amt-19-6415-2026

Chyzhykov gravimetric filter water Karl Fischer bias 2026 Research Square (preprint)

Impact of Mass Loading and Water Speciation on Precision and Artifacts in Gravimetric Measurements of Atmospheric Particulate Matter

Thirty filters (n=5 per fraction per site) plus five field blanks, weighed under PN-EN 12341:2024 conditioning, then stepwise thermal water release at 50, 120, 180 and 250 C. Between-filter variability was U-shaped in mass loading, minimal near 1330-1360 ug; PM_1 was the most variable fraction (CV up to 53%). Strongly bound water (180-250 C) made up 51-60% of filter water, survives standard conditioning, and adds a systematic +3% to +18% mass bias, largest for PM_1 at the steelworks. Limitation: 35 filters at two industrial sites; the bias is not yet resolved by composition or RH history, so transfer to urban background aerosol is untested. Relevance: high - every LCS calibration inherits the reference's error; a positive, composition-dependent water bias in gravimetric PM_1 and $PM_{2.5}$ propagates directly into co-location slopes.

doi: 10.21203/rs.3.rs-11157411/v1

Drone boundary-layer profiles Germany 2026 Atmos Chem Phys

Influence of irradiance and mixing layer height on the vertical trace matter distribution in the lower planetary boundary layer - drone-based investigation

Two 2.5-week summer campaigns: night profiles in the lowest 120 m showed multi-layer fine-scale stratification; O_3 and potential temperature were the most reliable mixing-height markers, and trace species outperformed several meteorological variables. Local sources and sinks affected in-layer distribution more than irradiance, but aerosol concentrations homogenised under strong radiation even at low mixing heights. Limitation: rural summer only; drone-borne particle sensors are themselves low-cost optical units with flow and RH artefacts that the abstract does not address. Relevance: high - directly tests when a ground node represents the mixed layer.

doi: 10.5194/acp-26-14417-2026

Dust lidar asymptotics 2026 Atmos Chem Phys

Asymptotic behavior of lidar scattering properties of absorbing dust aerosols across Rayleigh and geometrical-optics regimes: theory and implications

TAMUdust2020 and triaxial-ellipsoid calculations give shape-insensitive limits: Rayleigh S ~ x^-3, SSA ~ x^3, small depolarisation; geometric-optics depolarisation falls, SSA -> 1/2 and lidar ratio rises with strong absorption. Large, strongly absorbing dust can therefore mimic weakly depolarising classes and bias classifiers toward under-counting coarse and super-coarse dust. Limitation: theory only, no field test. Relevance: moderate - the same physics explains why optical PM sensors under-read coarse absorbing dust; the bias direction carries over.

doi: 10.5194/acp-26-14487-2026

Epple DLR tire-wear particle chassis dynamometer 2026 Atmosphere

Temperature and Inflation Pressure Effects on Airborne Tire Wear Particle Measurements on a Chassis Dynamometer

Two-level factorial (inflation 1.8/3.8 bar x chamber 11/35 C) on the DLR TRWP system, in two blocks of toe setting. Higher temperature raised cumulative MCPC response 3.6-fold (95% CI 1.8-7.2); higher pressure cut OPS number 18% (12-24%); OPS mass proxy fell ~17% with temperature. Toe >48 arcmin produced rising signals and irregular wear. Reference-condition MCPC response doubled over the campaign, so drift monitoring and randomised blocked run order are prerequisites. Limitation: one tire, one rig; the OPS mass proxy assumes density and refractive index; the ultrafine mode may be volatile, not wear. Relevance: moderate - non-exhaust PM metrology with explicit instrument drift; the same discipline applies to LCS.

doi: 10.3390/atmos17101011

Fluorescence-Raman-Mie lidar $PM_{2.5}$ profiles Beijing 2026 Atmos Meas Tech

Classification-constrained retrieval of $PM_{2.5}$ vertical profiles from fluorescence-Raman-Mie polarization lidar

Beijing lidar separates anthropogenic pollution aerosol from desert and mineral dust using particle depolarisation ratio plus fluorescence capacity, then applies an extinction-to-$PM_{2.5}$ regression only to the anthropogenic fraction. Date-separated near-surface validation (411 hourly samples, 69 days): RMSE 17.32, MAE 13.54, bias 3.30 $\mu g\,m^{-3}$; versus the unconstrained model RMSE -38.2%, MAE -45.0%, bias about -83%. Limitation: validation is at the surface only, so the vertical profile itself is unverified, and residual scatter of the empirical extinction-mass relation, not the classification, dominates uncertainty. Relevance: high - the same coarse-dust mass bias wrecks optical LCS; type-gated calibration is directly transferable to network QC during dust events.

doi: 10.5194/amt-19-6517-2026

HMAC message authentication vs LoRa telemetry poisoning 2026 Sensors (Basel)

Securing AI-Driven Systems for Environmental Sustainability in IoT Applications: A Lightweight Message-Authentication Scheme Against Telemetry Poisoning

Proposes an encryption-free HMAC-SHA-256 tag over device ID, timestamp, monotonic counter and payload for raw LoRa links, with a durable server-side counter high-water mark to reject forgeries and replays. Host-side reference tests: zero false accepts/rejects over 25,000 adversarial messages. In a ten-seed poisoning study on public environmental datasets, a value-based outlier defence raised $PM_{2.5}$ forecast RMSE by up to ~104% by discarding real extremes, whereas authentication held the clean baseline. Cost: payload 34 to 99 B, airtime ~2.2x. Limitation: never run on the ESP32-S3 hardware; energy unquantified. Relevance: moderate - smoke-event extremes are precisely what outlier filters in LCS pipelines wrongly remove.

doi: 10.3390/s26196276

He dust-storm PM_{10} deep learning 2026 Atmos Environ

Transport-aware spatiotemporal deep learning for accurate PM_{10} forecasts during dust storms

METADATA ONLY. No abstract at harvest. Title indicates a spatiotemporal deep-learning PM_{10} forecaster with explicit transport awareness, targeted at dust storms where station-only models lag the plume. Recorded for coverage; no metric is inferred. Relevance: moderate - upwind sensor placement is the obvious data complement to transport-aware models.

doi: 10.1016/j.atmosenv.2026.122409

Kalisa East Africa passive sampler network PAHs metals 2026 Environ Pollut

First Results from a Passive Air Monitoring Network across Five East African Countries for Polycyclic Aromatic Hydrocarbons (PAHs) and Metals

Ten sites, 30-day dry-season deployments everywhere and paired wet-season samples at four. PAH ring profiles and metal loadings were spatially heterogeneous; water-soluble metals were a small share of total. HYSPLIT put >90% of air masses from the S-SE, and adjacent cross-border sites saw near-identical air, so concentration differences are local-source driven; PCA components explained 46.5% and 19.9% of variance. Wet-season levels were lower. Screening risk was metal-dominated at some sites and PAH-dominated at Bukavu. Limitation: PUF-PAS uptake rates for particle-bound species are uncertain and site-specific, so concentrations are semi-quantitative; no PM mass. Relevance: high - electricity-free composition sampling is the natural complement to optical LCS mass networks in data-sparse Africa.

doi: 10.1016/j.envpol.2026.129361

MISR Research Product Algorithm 1.1 km AOD 2026 Atmos Meas Tech

The MISR Research Product Algorithm - producing global, self-consistent, pixel-level aerosol retrievals for the Multi-angle Imaging SpectroRadiometer

Automates the MISR research aerosol retrieval into a production algorithm (MRPA) at 1.1 km pixels versus 4.4 km for the standard product. Against AERONET over land, MRPA mid-visible AOD has roughly half the RMSE and a higher correlation than the standard algorithm, at the cost of somewhat higher bias; Angstrom exponent improves, especially at high AOD. Smoke regions retrieve small-medium spherical absorbing particles, dust regions large non-spherical weakly absorbing ones. Limitation: automation reduces particle-type sensitivity, SSA and shape are validated only qualitatively, and MISR's 9-day global revisit limits daily $PM_{2.5}$ use. Relevance: moderate - type-resolved AOD is a plausible covariate for compositional correction of LCS-satellite fusion.

doi: 10.5194/amt-19-6503-2026

Nobin Bangladesh GNN $PM_{2.5}$ 2026 PLoS One

Explainable spatiotemporal Graph Neural Networks: Multi-City $PM_{2.5}$ forecasting in Bangladesh with geographic graph topology validation

Graph-diffusion GRU on four years of daily multi-source data at 12 cities. Up to 3-day lead: MAE 8.85 $\mu g\,m^{-3}$, R2 0.666, beating non-graph baselines (FDR-corrected Diebold-Mariano $p<0.001$) and DCRNN when pooled across horizons. The useful result is negative for complexity: a fixed distance-adjacency graph beat adaptive, wind-informed and hybrid graphs, and occlusion-based influence tracked geographic adjacency ($r=0.928$). Limitation: 12 nodes is too sparse to discriminate graph topologies - transport structure that matters at 10-50 km scale cannot be resolved by stations hundreds of km apart. Relevance: high - a direct test of what graph priors a denser LCS network would need before learned adjacency pays off.

doi: 10.1371/journal.pone.0351928

Qian MAG-MISD probabilistic multi-station $PM_{2.5}$ forecasting 2026 Preprints.org (preprint)

MAG-MISD: A Mode-Specific Adaptive Graph Framework for Multiscale Interaction and Station Dependence in Probabilistic $PM_{2.5}$ Forecasting

Variational mode decomposition into five components, cross-mode Transformer per station, and a separate directed graph per mode (geographic, persistent adaptive and sample-conditioned dynamic edges) before fusion into conditional quantiles. Against the same-input VMD-iTransformer, RMSE fell 4.76%, 2.78%, 0.13% and 0.72% at 1, 6, 12 and 24 h, with better pinball loss and interval score and ~70% fewer parameters; the persistent adaptive relation drove most of the gain in ablation. Limitation: the improvement vanishes beyond 6 h, and decomposition of the full series before splitting risks leakage unless done causally - the abstract does not say. Relevance: moderate - station-graph forecasting transfers to LCS networks, where per-mode dependence could also flag drifting nodes.

doi: 10.20944/preprints202610.1533.v1

SEARO air-quality early-warning review 2026 Front Public Health

Prediction accuracy of available early warning systems for air pollution across WHO South-East Asia region: a systematic review

PROSPERO-registered review (to Feb 2026): 14 modelling studies from 839 records, PROBAST+AI risk-of-bias. Hybrid deep-learning forecasters reported the lowest RMSE (0.84-4.61 $\mu g\,m^{-3}$) and R2 up to 0.998 for $PM_{2.5}$; ML reached 98.3% categorical accuracy; CTMs cut bias ~70% only with satellite data assimilation. Most studies were at high risk of bias from absent external validation and overfitting; heterogeneity precluded meta-analysis. Limitation: an R2 of 0.998 for daily $PM_{2.5}$ is a leakage signature, and the review reports such figures without re-analysis. Relevance: moderate - operational forecast skill in the region is unproven; LCS networks could provide the out-of-sample validation these systems lack.

doi: 10.3389/fpubh.2026.1945362

Saharan fresh dust optics 2026 Atmos Chem Phys

Optical properties and shape-dependent retrievals of the imaginary part of the refractive index of freshly emitted Saharan dust

In-situ optics during active emission events (Sept 2019): SSA for $PM_{2.5}$ (PM_{10}) 0.95 (0.94) at 370 nm and 0.97 (0.96) at 660 nm; AAE to 2.5 (2.0) from iron oxides; MAE fell from 0.30 to 0.15 m2/g at 370 nm with size. The imaginary index was retrieved under prescribed shape and density. Limitation: the retrieval depends on the shape assumption it is titled after, and a single campaign. Relevance: moderate - mass scattering efficiency shifting with size is the physics behind optical sensors' coarse-dust under-response; these values are usable priors for a dust correction.

doi: 10.5194/acp-26-14531-2026

Schafer AERONET lunar AOD assessment 2026 Atmos Meas Tech

An assessment of lunar photometry in AERONET

CIMEL model-T lunar AOD at 492 sites since 2014 with an updated empirical correction to the ROLO lunar irradiance model. At all long-record sites, day-night AOD differences are typically within +/-0.01 at 440 nm and as good or better at longer wavelengths; in 3 h day/night transition windows under low, stable AOD, AOD, Angstrom exponent, water vapour and SDA fine/coarse AOD show statistically negligible differences. Limitation: agreement is shown only under stable low AOD - smoke and dust nights, where night AOD matters most, are untested, and lunar-phase coverage leaves gaps. Relevance: high - validated night-time AOD removes the daytime-only constraint on AOD-$PM_{2.5}$ fusion and gives LCS networks a night-time column reference.

doi: 10.5194/amt-19-6487-2026

Shai Nanozen RT vs gravimetric 2026 Ann Work Expo Health

Comparison of respirable dust concentrations between gravimetric weighing and real-time dust monitoring equipment in non-mining industries

20 paired task-based stationary samples at 11 industrial sites. No significant difference between gravimetric and raw real-time (Wilcoxon p=0.067) or density-adjusted (p=0.277); proportional bias grew with concentration but was absent below 1 mg/m3 (R2 0.06 and 0.02). Limitation: n=20, and the gravimetric reference is the instrument's own cassette downstream of the same inlet, so inlet losses and size-selection error are shared and invisible to the comparison; Wilcoxon non-significance at n=20 is not agreement. Relevance: moderate - the 1 mg/m3 cut is a usable operating limit, and the shared-inlet design is a pattern to avoid in LCS co-location.

doi: 10.1093/annweh/wxag086

Stutz Fairbanks inversion layers 2026 Atmos Chem Phys

Vertical distribution of pollution trapped by wintertime surface inversions in Fairbanks, Alaska, during the ALPACA experiment

Long-path DOAS profiles during ALPACA 2022 resolve polluted surface layers 20-40 m deep persisting for days, with SO_2 to 35 ppb, NO_2 60 ppb, HONO 2.5 ppb, HCHO 7 ppb and depleted ozone. PACT-1D, with mixing parameterised from surface temperature gradient and wind, reproduces the profiles and gives a 1-4 h residence time. Limitation: gas-phase only - particulate profiles, the health-relevant quantity in Fairbanks, are not retrieved - and one winter at one city. Relevance: high for siting - a 20-40 m layer means a node on a rooftop and one at breathing height sample different air masses under inversions.

doi: 10.5194/acp-26-14345-2026

Sylvester PurpleAir Philadelphia 2026 ACS EST Air

Validation of PurpleAir $PM_{2.5}$ Sensors in Temperate Urban Environments

Co-location in Philadelphia across summer and spring plus a one-year re-evaluation. Best performance came from the nationwide Barkjohn (2021) correction applied to the mean of the two channels' atm outputs: RMSE 1.64 $\mu g\,m^{-3}$, MBE 0.10 $\mu g\,m^{-3}$, R2 0.683 at 1 h and 24 h, ahead of locally derived corrections. RH source (on-board vs weather station) changed R2 by only 0.04. After a year in the field R2 fell to 0.567. Limitation: R2 0.68 at low temperate concentrations means RMSE is the honest metric and the R2 drop partly reflects a narrower concentration range in the later period, not drift alone - the abstract does not separate them. Applying Barkjohn, derived on cf_1, to atm outputs is off-label. Relevance: high - direct evidence that a national correction can beat local fitting, and that year-one degradation needs scheduled re-co-location.

doi: 10.1021/acsestair.5c00443

Taiwan transboundary PM_{10} forecast 2026 Environ Pollut

Forecasting transboundary PM_{10} pollution in northern Taiwan using upwind observations and transport timing

Upwind-transition-receptor framework with temporally separated training and evaluation for 1-4-day PM_{10} forecasts. Nested models attribute most added skill to distributed upwind observations; trajectory timing adds a smaller, receptor-dependent refinement. Best model (CatBoost with upwind data): RMSE 11.04 $\mu g\,m^{-3}$, 13.0% below persistence. Limitation: 13% over persistence at 1-4 days is modest, and upper-tail precision is limited by event scarcity - the episodes that matter are the ones the model sees least. Relevance: high - quantitative evidence that upwind observation density, not trajectory sophistication, buys forecast skill; a siting argument for upwind nodes.

doi: 10.1016/j.envpol.2026.129340

UAV rotor-wash bias in airborne pollutant sensing 2026 Sensors (Basel)

UAV-Induced Measurement Bias in Air Pollution Sensing: A Physics-Informed Framework for Characterization, Sensor Placement, and Correction

Actuator-disk momentum theory and literature constraints are combined into a four-zone rotor-wash contamination map, a sensor-placement rule and a two-stage bias correction for nine pollutants. Hover and 5 m/s co-location flights on a 21-inch quadrotor against a fixed reference showed bias of 3-10% in hover and 6-15% in forward flight across NO_2, CO, O_3, PM_{10} and $PM_{2.5}$, directionally matching predictions, but correlation fell to r = 0.44 in forward flight. Limitation: a single airframe, few flights and no wind tunnel - the authors concede qualitative rather than quantitative support. Relevance: high for vertical-profile LCS work - mounting position is a first-order error term and r = 0.44 is below any usable validation threshold.

doi: 10.3390/s26196178

Wildfire smoke onset detection fixed cameras 2026 Sensors (Basel)

Detecting Wildfire Smoke Onset from Fixed View Camera Networks

Reframes smoke detection as onset detection: a scene model learned from unlabelled fire-free video yields per-tile prediction residuals, read through four cues (residual magnitude, opacity accretion, motion coherence, temporal persistence), with detection delay and false alarms in the loss. On FIgLib the first alarm comes 1.52 min after ignition at 0.21 false alarms per camera-day, transferring to PyroNear-2025 and SKLFS-WildFire without labels. Limitation: ignition-to-alarm is benchmarked on curated sequences; night, fog and low-cloud performance is not reported. Relevance: low-moderate - an upstream trigger, not a PM measurement, but camera onset times could gate event-mode sampling and plume attribution in LCS networks.

doi: 10.3390/s26196020

Exposure assessment & modelling — 26 records

Carried from the daily issues — full summaries (26 of 26 records in this subtopic)

Chen Chengdu WSON partitioning 2026 Atmos Chem Phys

Simultaneous measurements of gas- and aerosol-phase water-soluble organic nitrogen in the winter urban atmosphere of Chengdu, China

Winter urban campaign: particulate WSON 2.3+/-1.4 ugN/m3, twice gaseous, rising with $PM_{2.5}$; particle fraction 0.70+/-0.14. WSON partitions into organic phase under warm-dry and into aerosol liquid water under cold-humid conditions; amines are mainly gaseous but shift to particles as aminium sulfate/nitrate during episodes. ALWC dominated partitioning over T, RH and pH; $NH4NO_3$ was the main ALWC contributor. Limitation: one site, one season, and thermodynamic S-curve confirmation is model-on-model. Relevance: moderate - ALWC is precisely what inflates optical PM readings at high RH, so the nitrate-ALWC link is a calibration covariate as well as a chemistry result.

doi: 10.5194/acp-26-14329-2026

Cheng BTH multi-source satellite $PM_{2.5}$ stacking 2026 Atmosphere

Machine Learning-Based Retrieval of $PM_{2.5}$ Concentrations Using Multi-Source Satellite Data in the Beijing-Tianjin-Hebei Region of China

BTH 2019-2021: RF, SVR, XGBoost and linear stacking on four predictor sets with an 80:10:10 split. Test R2 rose from 0.51-0.54 with AOD alone to 0.83-0.84 with all eight predictors; stacking best at R2 0.84, RMSE 12.4, MAE 9.7 $\mu g\,m^{-3}$. Southern/southwestern Hebei highest; winter > summer. Missing AOD limited spatial completeness. Limitation: a random split leaks spatial and temporal autocorrelation, so the test R2 overstates out-of-site skill; no spatial-block CV, and gap filling - the real problem - is left to future work. Relevance: moderate - AOD alone explains half the variance; ground nodes and meteorology carry the rest.

doi: 10.3390/atmos17101010

China 20 cities building green $PM_{2.5}$ 2026 Environ Pollut

City-size-dependent interactive effects of building and green-space on $PM_{2.5}$ pollution: Evidence from 20 representative Chinese cities

metadata only. No abstract at harvest. Title indicates interactions of building form and green space with $PM_{2.5}$ that depend on city size. Recorded for coverage. Relevance: low-moderate.

doi: 10.1016/j.envpol.2026.129344

DC 1 km WRF-SMOKE-CMAQ 2026 ACS EST Air

Neighborhood-Scale O_3, NO_2, and $PM_{2.5}$ in Washington, D.C. from a 1 x 1 km2 WRF-SMOKE-CMAQ Framework

1 km CMAQ for 2017: hourly O_3 r up to 0.7; NO_2 biased -14 to -29% from unresolved near-road gradients. NO_2's coefficient of variation exceeded O_3's and was more than three times $PM_{2.5}$'s; urban-core O_3 was 4-11% lower from titration. Moving from 4 to 1 km changed population-weighted exposure by only -1.5 to -3% for $PM_{2.5}$. Limitation: evaluation against a handful of AQS sites in one city cannot test intra-urban gradients - the very thing 1 km is meant to add. Relevance: high - quantifies that $PM_{2.5}$ is the spatially smoothest pollutant, so dense $PM_{2.5}$ networks buy less spatial information than NO_2 networks would.

doi: 10.1021/acsestair.6c00155

Delhi-NCR COVID transitions 2026 Sci Prog

Time series evidence on air quality transitions under urban shock during COVID-19 shutdown in Delhi-NCR, India

Ten regulatory stations across pre-, during- and post-COVID periods analysed with ARIMA, PCA, change-point detection and recurrence quantification: significant reductions and altered temporal dynamics during lockdown, with meteorology and secondary formation relatively more important. Limitation: no deweathering is described, so lockdown effects are confounded with seasonal meteorology; the lockdown literature on Delhi is saturated. Relevance: low.

doi: 10.1177/00368504261498073

Five megacities $PM_{2.5}$ O_3 trends meteorology 2018-2023 Atmosphere

Urban Air Quality Trends and Meteorological Influences Across Five Global Megacities, 2018-2023

Descriptive 2018-2023 comparison. Shanghai $PM_{2.5}$ fell from 43.5 to 27.0 $\mu g\,m^{-3}$ (2018-2022) then rose to 30.3 in 2023; Delhi stayed above 87 $\mu g\,m^{-3}$ throughout; Paris declined; Sao Paulo rose by 2023. $PM_{2.5}$ exceedance days: Delhi 2,025, Shanghai 1,881, Los Angeles 366. Temperature correlated negatively with $PM_{2.5}$ in Delhi ($r = -0.66$), sea-level pressure positively ($r = 0.65$); wind speed was negative everywhere. Limitation: pairwise correlations without meteorological normalisation, so trends conflate emissions and weather; exceedance counts summed over pollutants and stations are hard to compare. Relevance: low - context only.

doi: 10.3390/atmos17101021

Ghazaryan explainable ML annual $PM_{2.5}$ Europe Central Asia 2026 Preprints.org (preprint)

Explainable Machine Learning for $PM_{2.5}$ Prediction and Comparative Country-Level Risk Assessment in Europe and Central Asia

Fifty-nine country-year observations (ten cities, 2020-2025) with eleven predictors; six models under five-fold CV. Linear regression had the lowest RMSE and highest R2, Extra Trees the lowest MAE; SHAP attributed added information to development indicators beyond meteorology. The authors call the result exploratory. Limitation: 59 observations and 11 predictors with random-fold CV on autocorrelated city panels cannot support model ranking or attribution; numeric skill is not reported in the abstract. Relevance: low - illustrates why annual aggregate predictors are no substitute for measured spatial coverage.

doi: 10.20944/preprints202610.1559.v1

Gray-green infrastructure $PM_{2.5}$ MSPA 2026 Build Environ

Spatially Explicit Morphological Regulation of Gray-Green Infrastructure for $PM_{2.5}$ Mitigation in High-Density Cities: An Interpretable MSPA-Machine Learning Framework

metadata only. No abstract at harvest. Title indicates morphological spatial pattern analysis plus interpretable ML relating gray-green infrastructure to $PM_{2.5}$ in dense cities. Recorded for coverage. Relevance: low.

doi: 10.1016/j.buildenv.2026.115406

Hanoi $PM_{2.5}$-bound PCDD/Fs 2026 Atmos Pollut Res

Levels and contributing sources to $PM_{2.5}$-bound PCDD/Fs and dl-PCBs in ambient air of Hanoi

metadata only. No abstract at harvest. Title indicates measurement and source attribution of particle-bound dioxins, furans and dl-PCBs in Hanoi $PM_{2.5}$. Recorded for coverage. Relevance: low - trace organics require filter sampling.

doi: 10.1016/j.apr.2026.103235

Harbin ammonium $PM_{2.5}$ and precipitation sources 2026 Environ Pollut

From atmosphere to river: Tracing the origins and deposition of ammonium in both $PM_{2.5}$ and precipitation in Harbin

METADATA ONLY. No abstract at harvest. Title indicates source tracing of ammonium in $PM_{2.5}$ and precipitation in Harbin and its deposition to river systems. Recorded for coverage; no estimate inferred. Relevance: low - secondary inorganic aerosol source apportionment.

doi: 10.1016/j.envpol.2026.129335

Li Guangzhou 3D vegetation 2026 Atmosphere

Scale-Dependent Effects of Three-Dimensional Urban Vegetation on Land Surface Temperature and Air Pollutants: Synergy and Trade-Off Zones in Guangzhou, China

Field LAI in 570 quadrats showed MODIS LAI is uninformative (r=0.15), so 3-D greenness was proxied by canopy height x tree cover. MGWR at 1-5 km: one SD of vegetation volume associated with 1.3 C lower LST, 0.30 $\mu g\,m^{-3}$ lower $PM_{2.5}$ and 1.3 $\mu g\,m^{-3}$ lower NO_2; O_3 null citywide with offsetting local responses. Limitation: $PM_{2.5}$ is a 1 km product whose own predictors include land cover, so the vegetation coefficient is partly the model reading its inputs back. Relevance: low-moderate - 0.3 $\mu g\,m^{-3}$ is below the error floor of any LCS network, which is itself the point.

doi: 10.3390/atmos17101001

Lucknow Holika Dahan $PM_{2.5}$ 2026 Environ Monit Assess

Impact of Holika Dahan on ambient $PM_{2.5}$ and associated health risks in an urban environment of Lucknow, India

Pre-, during- and post-Holika Dahan nights at a residential and a commercial site: $PM_{2.5}$ rose 2.0x and 1.4x; burning tracers 1.2-2.7x; PCA/HCA identified festival biomass burning as the dominant source; AERMOD placed hotspots out to 1.2-1.5 km. Hazard indices stayed below USEPA limits. Limitation: two sites over three nights, and a chronic-dose hazard index applied to a single-night event is the wrong risk metric. Relevance: moderate - a short, intense, spatially bounded event of exactly the kind a dense network resolves and two filter sites do not.

doi: 10.1007/s10661-026-16041-2

Monsoon aerosol Pb isotopes rainwater offsets 2026 Atmos Environ

Monsoon-related variability of water-soluble aerosol Pb isotopes and systematic aerosol-rainwater offsets at Peng Chia Yu

METADATA ONLY. No abstract at harvest. Title indicates monsoon-resolved variation in water-soluble aerosol Pb isotope ratios and systematic offsets between aerosol and rainwater signatures. Recorded for coverage; no estimate inferred. Relevance: low - source apportionment input, not a monitoring method.

doi: 10.1016/j.atmosenv.2026.122425

Multi-scale $PM_{2.5}$ nonlinear time-series two metros 2026 Atmos Pollut Res

Deciphering multi-scale $PM_{2.5}$ temporal dynamics and cross-scale interactions in two climatically distinct metropolitan areas: insights from advanced nonlinear time-series analysis

METADATA ONLY. No abstract at harvest. Title indicates nonlinear multi-scale decomposition of $PM_{2.5}$ series and cross-scale interactions in two climatically distinct metropolitan areas (author affiliations suggest Vietnam/Taiwan). Recorded for coverage; no estimate inferred. Relevance: low-moderate - scale separation is how network redundancy and siting density are usually argued.

doi: 10.1016/j.apr.2026.103238

North Macedonia biomass PMF 2026 Environ Pollut

Biomass burning drives urban $PM_{2.5}$ across the Western Balkans: evidence from a six-city source apportionment study in North Macedonia

1,195 filters at six urban-background sites over two year-long campaigns. Campaign means 29.1-62.2 $\mu g\,m^{-3}$ (1.2-2.5x the EU annual limit); residential wood combustion and open/waste burning resolved as distinct factors everywhere, together 21-35% of $PM_{2.5}$, with heating-season RWC up to 33x non-heating. Biomass exceeded 50% of $PM_{2.5}$ on 83% of Skopje episode days; an undocumented winter signal in Strumica fits greenhouse boilers. Limitation: PMF factor naming relies on tracer interpretation, and RWC/OBB separation without levoglucosan isotopes or 14C is assumption-laden. Relevance: high - a wood-smoke regime with bimodal evening peaks to 93 $\mu g\,m^{-3}$, where optical sensor response to fresh smoke is the calibration problem.

doi: 10.1016/j.envpol.2026.129339

OA molecular composition predicts ROS yield MCR-VK 2026 Atmos Chem Phys

Linking molecular composition of organic aerosols to reactive oxygen species formation using an MCR-VK framework

Ambient fine PM from four Chinese megacities, a German semi-urban site and a boreal forest, plus chamber SOA, analysed by ultrahigh-resolution MS and H2O2/radical assays. Volume-normalised ROS rose with PM mass and dominates exposure in polluted cities, but mass-normalised ROS varied independently of loading and tracked the ratio of oxidised to unsaturated organic fractions (RF_OOC/RF_UOC); biogenic aerosol carried the highest per-mass yield. Limitation: correlational across few sites; cell-free ROS is a surrogate, and transition metals - a major ROS driver - are outside the framework. Relevance: moderate - mass-based LCS cannot see per-mass toxicity variation of this size; a case for pairing networks with periodic speciation.

doi: 10.5194/acp-26-14655-2026

Pourkiaei Santiago CMAQ EDGAR vs INEMA inventories 2026 Research Square (preprint)

High-Resolution Air Quality Modeling over Santiago, Chile: A Comparative Assessment of EDGAR and INEMA Anthropogenic Emission Inventories

Global EDGAR v8.1 (2022) vs regional INEMA v2 (2020) in CMAQ with three one-way nests to 1 km. Aggregate $PM_{2.5}$ emissions were similar but sectoral splits differed sharply; INEMA reproduced PM better, EDGAR daytime O_3. After scaling emissions to mean observations, source apportionment shifted between industrial combustion (EDGAR) and transport (INEMA), with building energy use important in both. DDM showed VOC-limited O_3 and sector-specific non-linear second-order $PM_{2.5}$ sensitivity to NOx. Limitation: calibrating emissions to the observed mean forces agreement and then attributes sources within that constrained state, so apportionment differences reflect inventory structure, not validated truth; performance statistics are not in the abstract. Relevance: moderate - 1 km CTM fields are the prior that dense LCS networks are fused with in Andean basins.

doi: 10.21203/rs.3.rs-10259950/v1

Pu Lima hybrid transfer-learning $PM_{2.5}$ 2026 ACS EST Air

A Hybrid Transfer Learning Approach for Ambient $PM_{2.5}$ Exposure Modeling in LMIC Contexts: A Case Study from Lima, Peru

Lima 2010-2023 daily $PM_{2.5}$ surface from an instance-weighted random forest that fuses reference and low-cost monitor data, satellite AOD and WRF-Chem. Sampling bias toward monitored districts is handled by geo-environmental similarity transfer learning that borrows weighted training instances from the US. Cross-validated R2 0.76; predicted mean $PM_{2.5}$ in mountainous districts fell from 34.19 to 29.23 $\mu g\,m^{-3}$ versus the unweighted model. Limitation: CV R2 is computed on the same biased monitor locations the method claims to correct, so the mountain-region gain is unverified by held-out sites there; how low-cost data were calibrated is not stated. Relevance: high - a template for using sparse LCS networks plus transfer learning in data-poor cities.

doi: 10.1021/acsestair.6c00125

Salem Riyadh $PM_{2.5}$ vegetation 2026 Environ Monit Assess

Spatiotemporal variation of $PM_{2.5}$ and its relationship with vegetation cover and meteorological factors during spring (2016-2022) in Riyadh, Saudi Arabia

16 stations, spring 2016-2022: seasonal mean $PM_{2.5}$ ranged from 94.6 (2018) to 55.0 $\mu g\,m^{-3}$ (2020), highest in southern and central Riyadh. Meteorological correlations were non-significant; the vegetation-$PM_{2.5}$ inverse association reached significance in one year only (rho=-0.539, p=0.031, n=16). Limitation: n=16 Spearman correlations per year cannot separate vegetation from land use, and dust events dominate spring variance; the one significant year out of seven is consistent with chance. Relevance: low-moderate - documents a dust-dominated regime where optical sensors' coarse-mode bias would govern any network.

doi: 10.1007/s10661-026-16057-8

Sao Paulo 2024 fire extremes 2026 J Environ Manage

Compound fire-meteorology extremes drove megacity air pollution during the 2024 South American fire season

Three multi-day 2024 episodes with daily $PM_{2.5}$ of 80-100 $\mu g\,m^{-3}$ in metropolitan Sao Paulo. Fire-off simulations missed the $PM_{2.5}$ and CO peaks; fire-attributable increments reached 17-21 $\mu g\,m^{-3}$ state-wide and a cumulative metropolitan burden of ~547 $\mu g\,m^{-3}$-day. Episodes differed in transport configuration, and the worst coincided with heat, dryness and weak ventilation. Limitation: all fire inventories underestimated the most extreme episode, so the attribution is a lower bound with unquantified error; a global model is coarse for a megacity receptor. Relevance: moderate - the inventory shortfall at the extreme is exactly where a dense surface network constrains attribution.

doi: 10.1016/j.jenvman.2026.131131

Temirzhanova Pavlodar snow aerosol 2026 Atmosphere

Geochemical Characteristics of the Pollution of Snow Cover with Aerosols in an Industrial Center (Pavlodar, North-Eastern Kazakhstan)

55 snow samples across industrial, buffer, non-industrial and background zones. Pb exceeded background 2-100-fold, with Pb, Sr and V peaking in industrial and buffer zones; PC1 loaded on rare-earth elements. Limitation: snowpack integrates wet and dry deposition over an unknown accumulation period, so it is a deposition map, not an air concentration map, and cannot be converted to inhalation exposure. Relevance: low - useful as a cheap spatial prior for siting composition samplers where no network exists.

doi: 10.3390/atmos17100995

Torii Kalimantan peat fire super-linear scaling 2026 Research Square (preprint)

Peat Burning Scales Super-linearly with Fire-Season Magnitude in the West Kalimantan-Sarawak Border Region, 2012-2025

Fourteen fire seasons of VIIRS detections against a peat map: peat-fire counts scale with total detections as a power law, exponent 1.96 (95% CI 1.57-2.36; R2 0.89), so doubling fire activity quadruples peat burning; robust to alternative predictors (2.36) and a Suomi-NPP-only subset (1.98). Peat share rises from <4% in La Nina years to 30-51% in El Nino years (ONI r 0.78), though ONI is collinear with season size. 24.3% of cells burning >=7 years gave 57.7% of cumulative FRP; scaling survives back-trajectory transport (1.59). Limitation: single-author preprint, detections not emissions, and no PM observations close the chain. Relevance: moderate - receptor-site sensors in Sarawak could test whether haze PM really tracks the squared term.

doi: 10.21203/rs.3.rs-11311408/v1

US school wildfire smoke trends 2026 Environ Res

Spatio-Temporal Trends in Wildfire Smoke Exposures at Schools and Childcare Centers in the United States and Implications for Child and Adolescent Health

METADATA ONLY. No abstract at harvest. Title indicates national spatiotemporal trends in smoke exposure at schools and childcare centres. Recorded for coverage; no estimate inferred. Relevance: high if the exposure surface is sensor-informed - school sensor deployments are the most common LCS siting in the US smoke literature.

doi: 10.1016/j.envres.2026.125883

Wang China $PM_{2.5}$ decade 2026 Atmos Pollut Res

From rapid decline to slower improvement: Decadal nonlinear evolution and regional predictive associations of $PM_{2.5}$ in China (2015-2024)

METADATA ONLY. No abstract at harvest. Title indicates a nonlinear trend analysis of Chinese $PM_{2.5}$ showing deceleration of the post-2015 decline and regional predictors. Recorded for coverage. Relevance: low-moderate - slower improvement means smaller year-on-year signals, which raises the bar on sensor drift control for trend detection.

doi: 10.1016/j.apr.2026.103234

Zhang Fusion-DeepRSM satellite $PM_{2.5}$ 2026 Environ Sci Technol

Fusion-DeepRSM for Satellite-Derived $PM_{2.5}$ Estimation and Precursor Sensitivity Prediction

Observation-constrained deep response-surface model fusing CTM runs, satellite products and ground $PM_{2.5}$. R2 0.872, RMSE 11.82 $\mu g\,m^{-3}$, and a 4.13 $\mu g\,m^{-3}$ reduction in rural overestimation, the known failure of station-trained ML. It also returns precursor-$PM_{2.5}$ response surfaces: for 2020-2023, meteorology offset part of emission-driven improvement (+1.2 vs -1.8 $\mu g\,m^{-3}$), and precursor change drove the Feb 2023 BTH rebound. Limitation: validation against stations that are themselves inputs; the rural bias correction is quantified on a CTM testbed, not against independent rural monitors - exactly where LCS would be needed. Relevance: high - a fusion architecture into which rural sensor nodes would slot as the missing constraint.

doi: 10.1021/acs.est.6c08137

Zhou condensation sink efficiency hygroscopicity 2026 Environ Sci Technol

Condensation Sink Efficiency of Atmospheric Particles Regulated by Their Chemical Compositions and Hygroscopicity

Defines a CS efficiency coefficient (epsilon = CS_eff/CS_calc, the latter assuming irreversible uptake) for gaseous H2SO4. Epsilon depends on composition and water uptake: dry pure organics scavenge less efficiently than inorganics, high RH raises epsilon for both, and mixed particles sit between, set jointly by composition and RH; CS efficiency is lower still for HIO_3. Including epsilon improves cluster-survival and sub-3 nm formation simulations and may explain intense NPF despite high CS in megacities. Limitation: epsilon values and their uncertainty are not given in the abstract, and field transfer is 'exploratory'. Relevance: low-moderate - NPF bursts drive number, not mass, so optical LCS miss them; CPC-equipped nodes would not.

doi: 10.1021/acs.est.6c07189

Cardiovascular & metabolic 7 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

Chen Lanzhou $PM_{2.5}$ components CVD 2026 Int J Environ Health Res

Acute cardiovascular mortality risks from $PM_{2.5}$ and its chemical components in an industrial valley city: evidence from Lanzhou, China

69,669 CVD deaths, 2015-2023. $PM_{2.5}$ lag 0: excess risk 2.06% per 10 $\mu g\,m^{-3}$ (0.66-3.48%), larger in men and under-65s. qgcomp/WQS attribute risk to PAHs and Tl/Pb/Mn in traffic-dominated Chengguan, and to nitrate/chloride/sulfate and PAHs in industrial Xigu. Limitation: components are not measured but predicted by a random forest with r>0.70, so the mixture weights inherit the predictor set of that model - district contrasts may reflect which inputs drove the prediction rather than toxicity. Relevance: moderate - a valley inversion city where source-specific nodes would test exactly these district contrasts.

doi: 10.1080/09603123.2026.2745894

Huang gut microbiota mediation air pollution metabolic syndrome 2026 iMetaOmics

The mediating roles of gut microbiota in the associations of air pollution and meteorological factors with metabolic syndrome

Gut microbial community clusters were derived and related to metabolic syndrome; health-associated clusters were dominated by Bacteroides and Prevotella. Long-term NO_2, $PM_{2.5}$ and temperature were associated with microbiota composition, and Megamonas and Megasphaera statistically mediated their associations with MetS. The abstract reports no sample size, effect estimates, mediated proportions or confidence intervals. Limitation: mediation from a single stool sample assumes exposure precedes the microbiome state and that diet, a strong common cause of both microbiota and MetS, is adequately adjusted - neither can be checked from the record. Relevance: low - mechanistic hypothesis generation; exposure resolution is not the limiting factor here.

doi: 10.1002/imo2.70099

MESA incident AF time-varying pollution 2026 Heart Rhythm

Association of Air Pollution Exposure and Risk of Incident Atrial Fibrillation: The Multi-Ethnic Study of Atherosclerosis

6,618 MESA participants (mean age 62), 16.5 y median follow-up, 1,055 incident AF events. Exposure came from the MESA Air hierarchical spatiotemporal models, which assimilate cohort-specific snapshot monitoring, entered as time-varying 2-y averages. Per IQR of NO_2 (13.5 ppb) AF hazard rose 29% (HR 1.29, 95% CI 1.00-1.67); $PM_{2.5}$, NOx and O_3 were weaker and directionally positive, with no estimates in the abstract. Limitation: AF ascertainment from hospitalisation and study ECGs misses paroxysmal disease, and the NO_2 CI touches the null; the $PM_{2.5}$ contrast within six metros is narrow. Relevance: high methodologically - this is the cohort where supplemental community monitoring was built to cut exposure error, and $PM_{2.5}$ still came out null-leaning.

doi: 10.1016/j.hrthm.2026.10.013

MESA long-term air pollution suPAR null 2026 Environ Epidemiol

Long-term ambient air pollution is not associated with circulating soluble urokinase plasminogen activator receptor

5,072 MESA participants (2000-2002 enrolment), ELISA suPAR, multivariable linear regression. No pollutant - $PM_{2.5}$, NO_2, NOx, O_3 or light-absorbing carbon - was associated with suPAR; the $PM_{2.5}$ CI excluded effects of 2% or more per 5 $\mu g\,m^{-3}$. With two prior null cohorts, the authors conclude suPAR is not a meaningful mediator of the PM-CVD link. Limitation: single cross-sectional suPAR measure, and the US contrast of the early 2000s is narrow relative to high-exposure settings, so a null at 5 $\mu g\,m^{-3}$ does not transfer to them. Relevance: low-moderate - a well-powered null worth citing against biomarker-mediation claims; the exposure model, not sensors, sets the contrast.

doi: 10.1097/ee9.0000000000000532

UKB arrhythmia NMR metabolome mediation 2026 Ecotoxicol Environ Saf

Large-scale plasma metabolic profiling identifies signatures and biomarkers associated with air pollution and arrhythmias: A prospective cohort study from UK Biobank

449,068 UK Biobank participants free of arrhythmia, median follow-up 16.3 y, 47,381 incident cases. All four pollutants were associated with arrhythmia, HR 1.015-1.257 per 10 $\mu g\,m^{-3}$ (range across pollutants; CIs not in the abstract). An elastic-net signature from 170 NMR metabolites mediated 11.2% of the $PM_{2.5}$ association (PM_{10} 5.2%, NO_2 7.4%, NOx 9.8%); lipid, ketone-body and amino-acid pathways recurred, ketone bodies most for intraventricular block. Limitation: single-baseline 2010 LUR exposure and single-time metabolomics make mediation cross-sectional in disguise; ketone bodies track fasting and diabetes status. Relevance: low-moderate for sensing - another UKB LUR result whose exposure contrast is too smooth to need dense monitoring.

doi: 10.1016/j.ecoenv.2026.120919

UKB women deprivation $PM_{2.5}$-NO_2 fatty liver 2026 Front Endocrinol (Lausanne)

Socioeconomic disadvantage, air pollution, and metabolic health among women: analyses of UK Biobank and US data

231,525 UK Biobank women: Townsend deprivation predicted FLI (beta 1.114, 95% CI 1.069-1.159) and FLI>=60 (OR 1.080, 1.076-1.084). Age-adjusted pollution associations vanished once deprivation entered the model (FLI beta -0.068, -0.137 to 0.001; FLI>=60 OR 0.999, 0.993-1.004). Across 46 US jurisdictions, +1 SD poverty meant +1.38 points female obesity after $PM_{2.5}$ adjustment. Limitation: adjusting for an area deprivation index that partly lies on the pollution pathway can over-adjust; a 2010 LUR score and state-level US $PM_{2.5}$ are coarse. The null is contextual, not causal. Relevance: low-moderate - shows how easily SES absorbs smooth LUR contrasts; finer exposure would be needed to separate them.

doi: 10.3389/fendo.2026.1948066

West Florida temperature x $PM_{2.5}$ CVD admissions 2026 J Am Heart Assoc

Joint Impacts of Ambient Air Pollution and Temperature on Cardiovascular Disease in West Florida

BayCare Health System admissions 2017-2024 analysed with distributed-lag GAMs and tensor-product temperature x $PM_{2.5}$ surfaces, copollutant-adjusted. Temperature dominated: MI risk in adults >=65 was lower at warmer days (RR75/25 0.92, 0.89-0.95), heart failure was U-shaped with heat risk in younger and Black patients (RR75/25 1.05, 1.02-1.09). $PM_{2.5}$ was weak; the largest signal was stroke in Black patients, RR75/25 1.02 (0.96-1.08). Limitation: a single-system admissions series assigns one area-wide monitor value to a coastal region with low, compressed $PM_{2.5}$, so near-null PM results partly reflect a small exposure contrast; subgroup multiplicity is unaddressed. Relevance: moderate - low-concentration subtropical settings are where denser networks would sharpen the contrast.

doi: 10.1161/jaha.126.049988

Neuro / mental health — 4 records

Carried from the daily issues — full summaries (4 of 4 records in this subtopic)

ALFA+ TRAP CSF GFAP 2026 Alzheimers Dement

Air pollution is associated with sex-specific longitudinal changes in astrocytic and neuroinflammatory processes linked to cognitive decline in Alzheimer's disease at-risk individuals

226 cognitively unimpaired middle-aged adults at elevated AD risk with baseline and 3-year CSF. Higher traffic-related exposure associated with longitudinal increases in GFAP and IL-6, mainly in women; GFAP partially mediated associations with worsening executive and attentional performance. Limitation: no effect sizes in the abstract; n=226 split by sex across four correlated pollutants and many biomarkers is a multiplicity problem, and single-address LUR estimates ignore time-activity. Relevance: moderate - CSF longitudinal biology is rare, but this is hypothesis-generating at this n.

doi: 10.1002/alz.71837

Jin wildfire carbonaceous ADRD 2026 medRxiv (preprint)

Carbonaceous Components Underlying Wildfire Smoke $PM_{2.5}$-Associated Alzheimer's Disease and Related Dementias in Older US Adults

39.7 million fee-for-service Medicare beneficiaries, 2005-2019, 8.4 million incident ADRD cases; stratified Cox. Per 1 $\mu g\,m^{-3}$, smoke $PM_{2.5}$ HR 1.029 (1.024-1.035) for ADRD and 1.015 (1.005-1.026) for AD; the smoke carbonaceous fraction HR 1.090 (1.077-1.103) for ADRD. Per IQR the non-smoke association is larger (ADRD 1.048 vs 1.009) because smoke exposure contrast is narrow. Attributable: 11,142 ADRD cases/year. Limitation: preprint; annual smoke partitioning is model-derived and its error is not propagated, and claims-based incident dementia carries diagnostic-intensity confounding that tracks rurality and fire-prone regions. Relevance: high - argues for composition-resolved smoke metrics that optical LCS cannot supply without co-located carbon measurement.

doi: 10.64898/2026.10.08.26364863

Khan Pakistan neurovascular review 2026 Curr Neurovasc Res

Air Pollution and Neurovascular Health in Pakistan: An Emerging Public Health Crisis in South Asia - A Narrative Review

Narrative synthesis of mechanisms - systemic inflammation, oxidative stress, endothelial dysfunction, BBB disruption, microglial activation, prothrombotic change - linking particulate exposure to ischaemic and haemorrhagic stroke and small-vessel disease, with the observation that Pakistan has no longitudinal, biomarker or exposure-linked stroke surveillance data. Limitation: no search strategy, no new data, no quantitative synthesis; the gap statement is the contribution. Relevance: low-moderate - a call for registry-linked exposure that presupposes the monitoring network Pakistan does not have.

doi: 10.2174/0115672026460820260929113419

Shanghai prenatal UFP postpartum depression 2026 Environ Pollut

Prenatal exposure to ultrafine particles and positive screening for postpartum depression: a prospective pregnancy cohort in Shanghai, China

2,647 pregnant women, two centres; 20.4% screened positive at 6-10 weeks postpartum. Per IQR, first-trimester UFP OR 1.252 (1.065-1.471) and whole-pregnancy UFP OR 1.104 (1.047-1.164); both survived adjustment for $PM_{2.5}$, which had no stable independent association. Stronger with male infants. Limitation: EPDS is a screen, not a diagnosis, and the UFP surface's spatial resolution and validation are not stated - UFP gradients decay over ~100-300 m from roads, so residential misclassification is likely large and non-differential only if traffic proximity is unrelated to deprivation, which in Shanghai it is not. Relevance: high - a second UFP-specific neuro signal this month that $PM_{2.5}$ mass does not carry, and that optical nodes cannot measure.

doi: 10.1016/j.envpol.2026.129330

Respiratory & allergic — 8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Guangdong COPD surveillance PM_1 $PM_{2.5}$ PM_{10} PAF 2026 Fundamental Research

Association between chronic respiratory diseases and the attributable risk of long-term exposure to atmospheric particulate matters among middle-aged and elderly Chinese population

COPD surveillance sample of 7,371 adults >=40 y (CRD 19.4%, COPD 10.0%). Each 10 $\mu g\,m^{-3}$ of PM was associated with higher prevalence of every CRD outcome, more so in rural residents and biomass users. PAF of CRD: PM_1 19.09% (95% CI 12.41-25.02), $PM_{2.5}$ 18.98% (12.90-24.90), PM_{10} 21.40% (13.84-27.67); COPD highest. Limitation: cross-sectional prevalence, so PAF here is causal only by assumption; $PM_1/PM_{2.5}/PM_{10}$ come from the same satellite-ML product and are near-collinear, so the three PAFs are one estimate three times; biomass fuel confounds rural contrasts. Relevance: low-moderate - PM_1 adds nothing distinct when derived from the same model.

doi: 10.1016/j.fmre.2025.04.013

Khan Gambia personal $PM_{2.5}$ pneumococcus 2026 medRxiv (preprint)

Longitudinal personal environmental exposure and Streptococcus pneumoniae colonisation in a cohort of children 5-14 years old in rural Gambia

134 children aged 5-14, 554 visits with paired personal exposure and culture. Geometric mean personal $PM_{2.5}$ 12.6 $\mu g\,m^{-3}$ (12.2-13.1), CO 0.16 ppm; colonisation 39.7%. Neither $PM_{2.5}$ nor CO was associated with colonisation after adjustment, including in within/between-child decomposition; between-child temperature was inversely associated (OR 0.74, 0.56-0.98). Limitation: personal $PM_{2.5}$ of 12.6 $\mu g\,m^{-3}$ in solid-fuel households is implausibly low for wood-smoke exposure and suggests the monitor or its wear compliance under-captured cooking peaks; a null under that exposure contrast is uninformative. Relevance: high - a clear case where personal-monitor validation, not the epidemiology, decides the result.

doi: 10.64898/2026.10.08.26364987

Kinzunga Kinshasa $PM_{2.5}$ respiratory 2026 Research Square (preprint)

Acute Respiratory Impacts of Fine Particulate Matter ($PM_{2.5}$) in Kinshasa: Epidemiological Evidence and Pulmonological Implications

13,049 adult admissions at one hospital, 2018-2022; mean daily $PM_{2.5}$ 46-50 $\mu g\,m^{-3}$. Per 10 $\mu g\,m^{-3}$ over lag 0-5, adjusted OR 1.01 (1.00-1.03), strongest at 35-64 years and in Tshangu and Funa. Limitation: a single-hospital catchment against a city-wide exposure series, and an estimate whose lower bound sits on the null; the sensor and model fusion step is described but its validation against any reference in Kinshasa is not. Relevance: high - one of the first African health analyses whose exposure series is built on low-cost sensors, which makes the uncorrected sensor error the main threat to the inference.

doi: 10.21203/rs.3.rs-11301519/v1

$PM_{2.5}$ constituents TB contacts 2026 Ann Am Thorac Soc

Differential effects of $PM_{2.5}$ constituents on tuberculosis infection and disease progression in 40,379 student close contacts

METADATA ONLY. No abstract at harvest. Title indicates a constituent-resolved analysis of $PM_{2.5}$ and latent infection and progression among 40,379 student close contacts of TB cases. Recorded for coverage; no estimate inferred. Relevance: moderate - a large contact cohort is an unusually clean design for an infection endpoint.

doi: 10.1093/annalsats/aaoag325

Sittitoon Thailand haze $PM_{2.5}$ symptoms inequality 2026 Front Public Health

Health inequalities in $PM_{2.5}$-related symptoms during Thailand's haze season: the role of education, sociodemographic, lifestyle, protective behaviors, and environmental factors

3,916 adults interviewed January-April 2025, mixed-effects logistic models with latent-class protective-behaviour phenotypes. Symptom prevalence: respiratory 80.7%, eye 52.8%, skin 48.7%, neurological 33.1%. Higher provincial $PM_{2.5}$ was independently associated with respiratory and dermatological symptoms; non-compliers (48.6%) and the neutral group had higher odds of respiratory and neurological symptoms than compliers (33.4%). Female sex, younger age, comorbidity, alcohol, urban residence and lack of village-health-volunteer contact also raised risk. Limitation: one satellite value per province per month is assigned to every respondent, so the $PM_{2.5}$ contrast is between-province and confounded with region; self-reported symptoms during a publicised haze season invite reporting bias; odds ratios are not given in the abstract. Relevance: high - northern Thailand has dense low-cost networks that could replace provincial satellite means with district-level exposure.

doi: 10.3389/fpubh.2026.1899785

Taiwan greenness pollution child asthma 2026 J Asthma Allergy

Association of Urban Greenness and Air Pollution on Asthma Visits Among Children and Adolescents in Industrial and Non-Industrial Cities in Taiwan

METADATA ONLY. No abstract at harvest. Title indicates an analysis of greenness and air pollution with paediatric asthma visits contrasting industrial and non-industrial Taiwanese cities. Recorded for coverage. Relevance: low-moderate.

doi: 10.2147/jaa.s614875

Trachalaki IPF $PM_{2.5}$ 2026 Ann Am Thorac Soc

Are Short-Term Increases in $PM_{2.5}$ Linked to Respiratory Hospitalizations in IPF?

METADATA ONLY. No abstract in PubMed, Europe PMC or Semantic Scholar at harvest; the title format suggests a research letter on short-term $PM_{2.5}$ and respiratory admissions in idiopathic pulmonary fibrosis. Recorded for coverage only - no result can be summarised, and none is inferred. Relevance: moderate if positive, because IPF is a small, identifiable cohort for which personal or home-sensor exposure is feasible.

doi: 10.1093/annalsats/aaoag317

Yoon pollutant asthma immune mechanisms review 2026 Allergy Asthma Immunol Res

Impacts and Immune Mechanisms of Pollutant-Mediated Asthma and Airway Inflammation

Narrative review of epithelial injury, barrier disruption, oxidative stress and innate signalling (NF-kB, MAPK, PI3K/Akt, inflammasome) driving T2, non-T2 and mixed asthma phenotypes; pollutant-specific arms include PM-induced NLRP3 activation, O_3 barrier disruption, DEP-driven IL-17A and pollutant-allergen co-exposure. Acknowledges that experimental doses and compositions do not map onto real-world exposure. Limitation: non-systematic, no new data, and no dose metric that a measurement study could target. Relevance: low - background for mechanism sections.

doi: 10.4168/aair.2026.18.5.640

Reproductive & developmental — 6 records

Carried from the daily issues — full summaries (6 of 6 records in this subtopic)

Guangdong CHD $PM_{2.5}$ constituents 2026 Environ Pollut

Differential Impacts of Fine Particulate Matter Constituents on Congenital Heart Disease across Critical Periconceptional Windows

Guangdong CHD Registry 2004-2022: 12,680 cases, 10,266 controls, multilevel logistic regression by exposure window. Per-IQR increments of $PM_{2.5}$ mass and constituents raised CHD odds 9-16%, strongest in the preconception window and attenuating later; BC and nitrate stayed associated during pregnancy. Conotruncal defects carried the largest ORs (1.14-1.36 per IQR). Mixture models put sulfate and BC first preconception, BC and nitrate thereafter. Limitation: constituents are modelled and mutually collinear, so 'dominant constituent' rankings are mixture-model artefacts as much as biology; registry controls and the higher-SES amplification hint at ascertainment differences. CIs absent from the abstract. Relevance: moderate - constituent-resolved windows need speciation, which optical LCS cannot supply; BC from multi-wavelength nodes is the tractable piece.

doi: 10.1016/j.envpol.2026.129345

Hyperemesis gravidarum PM_{10} temperature 2026 Front Public Health

Seasonal ambient temperature and particulate matter as discriminators of hyperemesis gravidarum severity: a retrospective comparative observational study

211 HG patients: temperature (AUC 0.799) and PM_{10} (AUC 0.754) discriminated severe from non-severe ketonuria; SO_2 did not. The PM_{10} association attenuated after adjustment for temperature. Limitation: hospital series with no control group, ecological exposure, and PM_{10} and temperature are seasonally collinear - the attenuation says PM_{10} was largely a season marker. Relevance: low.

doi: 10.3389/fpubh.2026.1957434

Liang CHD circulating proteins 2026 Nat Commun

A nested case-control study of air pollution and congenital heart defects via circulating proteins

Maternal periconception $PM_{2.5}$, PM_{10} and NO_2 were each associated with CHD, NO_2 most strongly; supervised sparse PLS identified pollutant-specific protein signatures in maternal and umbilical cord serum that tracked exposure and CHD, implicating FGF-4, VEGF-D, MSP and angiogenin. Limitation: the abstract gives no effect sizes or n; cord serum is sampled at delivery, months after cardiogenesis, so a cord-protein signature cannot be the mechanism of a first-trimester malformation, and supervised signature selection on the same outcome inflates association. Relevance: moderate - periconception exposure windows are short, which favours high-resolution residential estimates. Held from the 9 Oct issue.

doi: 10.1038/s41467-026-77623-9

Ni Wuhan birth cohort air pollution neurodevelopment target trial 2026 Environ Sci Technol

Early Life Ambient Air Pollution, Sex Steroids and Neurodevelopment: Target Trial Emulation in a Prospective Cohort

3,575 mother-child pairs (2014-2019), cord sex steroids in 897. Most pollutants were inversely associated with Bayley scores across windows, strongest in the first postnatal year. Androstenedione mediated 5-12% of some preconception/prenatal associations with PDI, but the IPW indirect effects lost significance. Target trial emulation of China's 2013-2017 Action Plan: children in the later policy period scored 7.27 (MDI) and 6.41 (PDI) points higher. Limitation: the policy contrast is a birth-period comparison, so secular changes in Bayley administration, parental education and care confound it; per-pollutant effect sizes and CIs are not in the abstract. Relevance: moderate - the postnatal window being most sensitive argues for exposure assigned to the infant's home, where indoor low-cost monitors outperform residential-grid estimates.

doi: 10.1021/acs.est.6c09006

Pakistan cooking fuel undernutrition 2026 Int J Environ Health Res

Association between household fuel use and childhood undernutrition in Pakistan: evidence from a National survey

4,224 under-fives; 45.4% with anthropometric failure, 37.7% stunted. Polluting vs clean fuels: CIAF OR 1.16 (1.04-1.31), stunting 1.16 (1.03-1.31), wasting 1.13 (1.00-1.29), underweight 1.17 (1.04-1.32); dung strongest. Limitation: fuel type is a wealth marker as much as an exposure, and with no measured PM the residual socioeconomic confounding is plausibly the size of the effect; cross-sectional. Relevance: moderate - the standard HAP proxy, and a reminder of why personal-monitor substudies inside DHS-type surveys are needed.

doi: 10.1080/09603123.2026.2749055

Placental angiogenic biomarkers review 2026 Reprod Toxicol

Exposure to environmental toxicants during pregnancy and placental angiogenic biomarkers: An integrative review

Whittemore-Knafl integrative review: 19 human studies after screening. Air pollutants predominantly showed an anti-angiogenic pattern - higher sFlt-1 or sFlt-1/PlGF and lower PlGF - as did PFAS, Cd and Cu; 14 of 19 studies reported at least one association in that direction. Limitation: 19 heterogeneous studies across four toxicant classes, no pooling or bias assessment, and 'at least one association' across multiple biomarkers and windows is a low bar. Relevance: low-moderate - plausible mechanism for PM-preeclampsia links, not evidence of it.

doi: 10.1016/j.reprotox.2026.109377

Mechanistic toxicology 15 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (15 OF 15 RECORDS IN THIS SUBTOPIC)

Chen toluene soot toxicity 2026 Atmos Environ

Oxidative toxicity and cytotoxicity of toluene combustion soot particles mediated by their microstructure

METADATA ONLY. No abstract retrievable at harvest. From the title: soot generated from toluene combustion, with oxidative potential and cytotoxicity attributed to particle microstructure (likely graphitisation/nanostructure order). Recorded for coverage; no result is inferred. Relevance: low-moderate - if microstructure rather than mass sets toxicity, mass-based optical sensing misses a toxicity axis entirely.

doi: 10.1016/j.atmosenv.2026.122407

Comis Turin oxidative potential 2026 Atmosphere

Oxidative Potential and Elemental Fingerprints of Urban and Suburban Particulate Matter in Turin

32 PM_{10} filters (urban traffic vs rural background, 2020-2021, min and max days of four months). Urban OP-AA 0.12-0.60 vs suburban 0.10-0.38 nmol AA/min/m3; Cu was the most active single metal. Total and phosphate-buffer bioaccessible metal fractions were compared. Limitation: filters were selected at monthly concentration extremes, so OP-mass relationships are inflated by design, and n=32 cannot support source attribution of OP. Relevance: low-moderate - OP remains unmeasurable by optical nodes; the Cu dominance points to brake-wear tracers as the practical proxy.

doi: 10.3390/atmos17101000

EPFR size-resolved PM 2026 Environ Toxicol Chem

Size-Dependent Characteristics, Exposure Risk, and Formation Mechanisms of Environmentally Persistent Free Radicals in Atmospheric Particulate Matter

EPFRs were carbon-centred radicals adjacent to oxygen groups; $PM_{2.5}$ had the highest mass-normalised concentration (8.05e16 spins/g) and g-factor (2.00337). Cigarette-equivalent exposure was 0.326/day for $PM_{2.5}$ and 0.346/day for PM_{10}. Laboratory surrogates showed smaller particles generate more EPFRs owing to larger specific surface. Limitation: the cigarette-equivalent model is a communication device without dosimetric basis, and site/season details are absent from the abstract. Relevance: low-moderate - another surface-chemistry toxicity axis that mass sensing cannot see.

doi: 10.1093/etojnl/vgag274

Edwards macrophage superoxide SOA biomass plastic burning PM 2026 ACS Omega

Superoxide Release from Macrophages upon Exposure to Secondary Organic Aerosols, Biomass, and Plastic Burning Particulate Matter

Macrophages exposed to PM suspensions, superoxide followed for 4 h by chemiluminescence plus EPR; an activation threshold is defined as the dose of a clear superoxide rise. Limonene SOA gave the largest release (~32 uM) but the highest threshold (300 ug/mL); naphthalene and toluene SOA gave 2.5-7 uM with thresholds of 0.02-0.1 ug/mL. Biomass PM had no clear threshold. Plastic-burning PM ranged widely: PET highest output (~12 uM), PS/LDPE/HDPE thresholds ~0.01 ug/mL, PVC and PET 50-100 ug/mL. Limitation: suspension doses in ug/mL do not map onto deposited alveolar doses, and a single cell line under submerged exposure omits surfactant and epithelial interaction. Relevance: moderate - per-mass oxidative potency varies by four orders of magnitude across sources, which mass-only sensors cannot register.

doi: 10.1021/acsomega.6c06562

Gut-liver axis pollutants MASLD review 2026 Int J Mol Sci

The Role of the Gut-Liver Axis in the Development of Liver Diseases Induced by Selected Environmental Pollutants: A Narrative Review

Narrative review covering heavy metals, PM and endocrine disruptors (chiefly BPA) as drivers of gut dysbiosis and barrier failure, with LPS translocation into the portal circulation activating hepatic TLR4/NF-kB, oxidative stress and steatosis; EDCs additionally act on hepatic nuclear receptors. Limitation: PM is one of three pollutant classes and the review does not separate inhaled from ingested particle routes or give dose context, so the PM-specific gut-liver claim rests on a thin, mostly rodent literature. Relevance: low for sensing - flags an extrapulmonary endpoint that cohort studies with dense exposure surfaces could test.

doi: 10.3390/ijms27198581

Houston UFP PAH deposition 2026 Air Qual Atmos Health

PAH-associated health risk from ultrafine particles: an urban field study

Three Houston sites: UFP respiratory deposition measured directly with the Mobile Aerosol Lung Deposition Apparatus, combined with size-segregated PAH analysis to estimate deposited PAH mass. Both carcinogenic and non-carcinogenic risks were within acceptable levels. Limitation: no numbers in the abstract; PAH risk is one narrow axis of UFP toxicity, so 'acceptable' does not translate to UFP being benign. Relevance: moderate - measured deposited dose rather than ambient concentration is the right exposure metric and is rarely available.

doi: 10.1007/s11869-026-02118-x

Inhaled polystyrene microplastics pulmonary fibrosis review 2026 Front Immunol

Polystyrene-associated pulmonary fibrosis: a summary based on the available evidence and prospects

Reviews sources, deposition and pulmonary effects of inhaled polystyrene microplastics and proposes, as a working model, convergence on coupled Wnt/beta-catenin and TGF-beta1/Smad2/3 signalling driving EMT, myofibroblast differentiation and matrix remodelling. Each element is graded as direct PS-MP evidence, extrapolation from other particles, or canonical fibrosis biology - an unusually honest structure. Limitation, stated by the authors: no study tests the full cascade in one inhalation model; non-physiological doses and pristine monodisperse spheres dominate. Relevance: low-moderate - airborne microplastics sit inside optical PM size bins but are invisible as a class to LCS; exposure assessment remains filter-and-spectroscopy work.

doi: 10.3389/fimmu.2026.1953398

Lung-on-a-chip pollutant toxicity review 2026 Front Bioeng Biotechnol

Microfluidic lung-on-a-chip for evaluating inflammation and toxicity from environmental pollutants

Review of microfluidic lung-on-a-chip platforms - air-liquid interface, cyclic stretch, perfused endothelium - for pollutant toxicology, covering PM, pathogens, VOCs and emerging nanomaterials, with advantages in biomimicry and real-time readout and the obstacles of chip standardisation, operational complexity and subsystem integration. Limitation: no quantitative synthesis; dose delivery (deposited mass per area) is the variable that decides translatability and is not addressed in the abstract. Relevance: low - only connects to monitoring if chips are dosed at measured ambient deposition rates.

doi: 10.3389/fbioe.2026.1942916

Mouth-throat replica deposition 2026 J Aerosol Sci

The influence of model selection and flow rate on aerosol transport in replica models of the mouth and throat: An experimental study

METADATA ONLY. No abstract at harvest. Title indicates an experimental comparison of mouth-throat replica geometries and flow rates for aerosol transport. Recorded for coverage. Relevance: low-moderate - dosimetry inputs.

doi: 10.1016/j.jaerosci.2026.106906

$PM_{2.5}$ HCC miR-25 2026 Transl Cancer Res

$PM_{2.5}$ promotes migration of hepatocellular carcinoma by targeting miR-25

$PM_{2.5}$ lowered E-cadherin, raised vimentin and miR-25 and increased MHCC97H migration; miR-25 mimics amplified and inhibitors attenuated the effect; mouse models showed enhanced tumour growth and invasion. Limitation: single cell line, unstated doses, and a single-miRNA mechanism for a pleiotropic exposure. Relevance: low.

doi: 10.21037/tcr-2026-0728

$PM_{2.5}$ NLRP3 inflammasome lung cancer review 2026 Int J Mol Sci

Possible Links Between $PM_{2.5}$-Induced NLRP3 Inflammasome Activation and Its Role in Lung Cancer Progression and Drug Resistance

Narrative review positioning $PM_{2.5}$-driven ROS as the trigger for NLRP3 inflammasome assembly, caspase-1 activation and IL-1beta/IL-18 maturation, sustaining chronic pulmonary inflammation in lung carcinogenesis. It then hypothesises that NLRP3 signalling contributes to lung-cancer drug resistance and proposes NLRP3 as a prognostic marker and target, noting NLRP3 inhibitors are in trials for other inflammatory diseases but none in oncology. Limitation: the drug-resistance link is explicitly hypothetical and assembled from heterogeneous in vitro doses; no systematic search or grading. Relevance: low for sensing - no exposure metric beyond mass $PM_{2.5}$.

doi: 10.3390/ijms27198931

$PM_{2.5}$ thyroid vitamin C rats 2026 Ecotoxicol Environ Saf

Vitamin C attenuates $PM_{2.5}$-induced thyroid dysfunction and mitophagy dysregulation by rebalancing ER-mitochondrial Ca2+ dynamics

Female rats and FRTL5 thyrocytes: $PM_{2.5}$ impaired thyroid function and histology, triggered ER stress, remodelled MAM proteins (IP3R1, GRP75, VDAC1), caused mitochondrial Ca2+ overload, ROS, depolarisation, PINK1/Parkin changes, GPX4 loss and Fe2+ accumulation; 4-PBA, 2-APB, Mdivi-1 and vitamin C each mitigated damage. Limitation: doses not given in the abstract and likely supra-environmental. Relevance: low - mechanism for a thyroid signal that the epidemiology (13 Oct fire-PM thyroid cancer) has not yet tied to function.

doi: 10.1016/j.ecoenv.2026.120920

Time-restricted feeding $PM_{2.5}$ adipose metabolism 2026 J Nutr Biochem

Metabolic Reprogramming by Time-Restricted Feeding Confers Resistance to $PM_{2.5}$-Induced Metabolic Disturbances in Adipose Tissue

METADATA ONLY. No abstract at harvest. Title indicates an experimental study in which time-restricted feeding attenuated $PM_{2.5}$-induced metabolic disturbance in adipose tissue. Recorded for coverage; no estimate inferred. Relevance: low for sensing - a dietary effect modifier, of interest only if human chrono-nutrition data are later paired with personal exposure.

doi: 10.1016/j.jnutbio.2026.110521

Yoon PRL-3 ezrin airway epithelium particulate matter 2026 Respir Res

PRL-3-sensitive cytoskeletal signaling mediates early airway epithelial responses to particulate matter

Phosphoproteomics of human bronchial epithelial cells under non-cytotoxic PM exposure, with pharmacological and genetic PRL-3 (PTP4A3) suppression and an acute mouse airway exposure model. PM rapidly and transiently upregulated PRL-3 and relocalised it to the membrane, with PRL-3-sensitive ezrin dephosphorylation in vitro and in vivo. Blocking PRL-3 restored ezrin phosphorylation and attenuated EMT-associated changes and MUC5AC upregulation, independent of Akt, ERK and mTOR, and reproducibly across UPM, DEP and PM_{10}. Limitation: doses are described as non-lethal but not related to deposited human doses; acute exposure says nothing about chronic remodelling. Relevance: low for sensing - a candidate early-effect pathway shared across PM types, i.e. not composition-specific.

doi: 10.1186/s12931-026-03834-9

Yu alveolar organoid $PM_{2.5}$ 2026 Front Toxicol

Differential toxicity of water- and DMSO-extractable $PM_{2.5}$ fractions in human alveolar organoids

Human alveolar organoids exposed to extracted $PM_{2.5}$ showed dose- and time-dependent morphological damage, more severe for the DMSO-extractable (organic) fraction, with ROS rise, cell death and a sustained fall in AT2 cells plus a small AT1 increase suggesting AT2-to-AT1 transdifferentiation. Limitation: doses are not reported in deposited-mass terms, so relevance to ambient exposure is unknowable, and extraction solvents select chemistry rather than particles. Relevance: low - model development. Held from the 9 Oct issue.

doi: 10.3389/ftox.2026.1943932

Occupational & indoor — 10 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 10 RECORDS IN THIS SUBTOPIC)

Creosote railroad switch PAH workers 2026 Ann Work Expo Health

Assessing PAH exposure during construction work of railroad switches using creosote-impregnated ties

Workers assembling railroad switches from creosote ties: breathing-zone sum of 16 PAHs 120,000-280,000 ng/m3 on a single sampled shift; skin pyrene 0.31-140 ng/cm2, highest on palms and calves; urinary 1-HP rose across the week (1.9-17 umol/mol creatinine), most above the UK guidance value by day 4. A PBTK model attributed uptake mainly to the dermal route, and pyrene under protective clothing implies fabric penetration. Limitation: one air-sampling shift, small crew, and the particle/vapour split is not reported, so the inhalation share rests on the model. Relevance: low for ambient sensing - a reminder that biomarker-based particulate PAH exposure is not reducible to air concentration.

doi: 10.1093/annweh/wxag088

Dental scaling bioaerosols 2026 J Hosp Infect

Divergent bacterial signatures in dental clinic bioaerosols: a real-field investigation

Ultrasonic scaling caused a large surge in sub-micron particle counts, but culturable bacteria concentrated in the 1.1-3.3 um stages. Airborne bacterial communities differed from PPE (ANOSIM R=0.633, p=0.001) and were indistinguishable from non-touch surfaces; oral pathogens were rare in air. Limitation: single clinic and procedure type, small sample count not given. Relevance: moderate - a clean demonstration that particle number from an optical counter is not a proxy for biological load; mirrors the 4 Oct suction study.

doi: 10.1016/j.jhin.2026.10.003

Global IAQ open standards framework 2026 Build Environ

A unified framework for global indoor air quality: transitioning from guidelines to actionable open regulatory standards

METADATA ONLY. No abstract at harvest. Title indicates a proposal to move indoor air quality from guideline values to open, enforceable standards. Recorded for coverage. Relevance: moderate - enforceable indoor standards would make low-cost indoor PM monitors compliance instruments, with the performance-certification burden that implies.

doi: 10.1016/j.buildenv.2026.115384

Indoor NPF from outdoor O_3 2026 Environ Sci Technol

Elevated Outdoor O_3 Can Sustain New Particle Formation and Growth Indoors

In an authentic office in a high-O_3 region, mechanical ventilation supplied enough O_3 that indoor new particle formation was VOC-limited, not O_3-limited; limonene addition drove growth at 20-100 nm/h, one to two orders above outdoor rates, with meaningful mass from grown particles. Limitation: one building, injected limonene rather than real product use, and no occupancy. Relevance: high - indoor UFP and secondary mass from cleaning products are invisible to $PM_{2.5}$-only indoor sensors until particles grow past ~300 nm.

doi: 10.1021/acs.est.6c07592

LiFePO4 workers cardiac autonomic 2026 Occup Environ Med

Association between occupational LiFePO4 exposure and cardiac autonomic function: a cross-sectional study of cathode material production workers

METADATA ONLY. No abstract at harvest. Title indicates a cross-sectional study of battery-cathode workers' LiFePO4 particle exposure and cardiac autonomic function. Recorded for coverage. Relevance: moderate - battery-material dust is an emerging occupational particle class with little exposure data.

doi: 10.1136/oemed-2025-110695

Office air purifier field CADR 2026 Build Environ

Air purifier performance in an occupied office: Field CADR and effectiveness from a reconstructed device-off condition

METADATA ONLY. No abstract at harvest. Title indicates field estimation of purifier clean-air delivery rate in an occupied office using a reconstructed device-off counterfactual. Recorded for coverage. Relevance: high if the counterfactual is sensor-derived - field CADR is how purifier-trial exposure contrasts should be verified.

doi: 10.1016/j.buildenv.2026.115390

Portable air purifiers airborne bacteria AMR offices schools 2026 Front Public Health

Response of culturable bacteria and their resistance to air purification across compartments

Offices in Valencia and classrooms in Dublin sampled before and after portable air purifier deployment; 341 isolates, MALDI-TOF identification, susceptibility testing and resistance-gene PCR. Office airborne isolates fell from 41 to 5; school airborne isolates fell from 102 to 47 and resistant prevalence from 43% to 21%. Surface and human-associated communities shifted less, and human-associated samples became Staphylococcus-dominated with more non-susceptible isolates. Limitation: single pre/post snapshots without control rooms or particle counts, culture-only recovery, small isolate numbers. Relevance: low-moderate - purifier trials rarely log bioaerosol and PM together; co-located OPCs would let isolate counts be normalised to removed particle number.

doi: 10.3389/fpubh.2026.1945592

Simkovich LPG stove RCT quality of life 2026 Indoor Air

Cooking With Liquefied Petroleum Gas or Biomass and Quality of Life: Results From a Randomized Controlled Trial

180 women primary cooks aged 25-64 randomised to an LPG stove, 12 months of free fuel and behavioural reinforcement vs usual cooking; 179 completed. ITT at 12 months: SF-36 mental 48.3 vs 50.9 (p=0.59), physical 53.1 vs 47.6 (p=0.08), EQ-5D VAS 67.0 vs 71.7 (p=0.05); no personal $PM_{2.5}$-QOL exposure-response. More intervention women felt financially better off (42% vs 13%) and reported higher income (33% vs 9%), both p<0.001. Limitation: QOL is a secondary outcome in a trial powered for something else, unblinded, and the fuel subsidy itself transfers income, so the financial result is partly mechanical. Relevance: moderate - personal $PM_{2.5}$ monitoring was available but the QOL signal was absent; intervention trials remain the strongest design for testing whether measured reductions matter.

doi: 10.1155/ina/9936046

Uberlandia street vendors $PM_{2.5}$ micronuclei panel 2026 Int J Hyg Environ Health

Urban air pollution and genotoxic damage in street vendors: a longitudinal repeated-measures panel study

238 men over four campaigns spanning rainy and dry seasons: 116 outdoor street vendors in the city centre vs 122 indoor formal vendors in a reference area. PM concentrations and MN frequency were both consistently higher in the centre. Poisson GEE: per 1 $\mu g\,m^{-3}$ $PM_{2.5}$, MN IRR 1.024 (1.013-1.034) rainy and 1.026 (1.019-1.034) dry; no season interaction (p=0.378). Limitation: exposure is fixed-point area monitoring, not personal, and is collinear with occupation group, so the $PM_{2.5}$ slope partly encodes street vs indoor work (traffic gases, PAHs, sun, smoking patterns). Relevance: moderate - seven fixed points is exactly the density at which a small low-cost array could convert an ecological contrast into personal-scale exposure.

doi: 10.1016/j.ijheh.2026.114932

Walter neighbourhood air exchange inhaled dose children 2026 J Expo Sci Environ Epidemiol

Neighborhood-scale disparities in air exchange rates and inhaled pollution dose among school-aged children

METADATA ONLY. No abstract at harvest. Title indicates neighbourhood-scale modelling of housing air exchange rates and the resulting inhaled pollutant dose among school-aged children, framed as a disparity analysis (Northeastern University group). Recorded for coverage; no estimate inferred. Relevance: high if infiltration is parameterised from data - indoor/outdoor LCS pairs measure exactly this ratio.

doi: 10.1038/s41370-026-00989-x

Burden, policy & mitigation — 18 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (18 OF 18 RECORDS IN THIS SUBTOPIC)

Abu Dhabi PM burden 2008-2022 2026 Risk Anal

Ambient Air Pollution Health Impacts in High-Income Countries, Part II: A Case Study in the United Arab Emirates

Attributable deaths from anthropogenic PM and ozone in Abu Dhabi rose from 231 (90% UI 59-427) in 2008 to 1,249 (507-2,169) in 2019-2022, despite air quality holding roughly steady while population doubled. The increase reflects population growth, better health records and, mainly, steeper concentration-response functions. Limitation: a fivefold change driven largely by swapping CRFs is a change in assumptions, not in burden; the 'anthropogenic' split in a dust-dominated desert regime is itself model-dependent. Relevance: moderate - shows how far CRF choice swamps measured trends, which bounds what better monitoring alone can change.

doi: 10.1111/risa.70368

Air-health web atlases review Italy 2026 Epidemiol Prev

PNC project 'Atlante Aria e Salute': exploratory web review and quality assessment of open-access atlases on the health effects of air pollution

Protocol-driven review of public air-and-health atlases (Sep-Dec 2024): 1,000 websites screened, 15 potentially relevant, 8 included (3 governmental, 3 academic, 2 foundations). All report $PM_{2.5}$ and/or PM_{10}; NO_2 and O_3 are less common; all but one use annual averages, six give dose-response estimates (mostly all-cause long-term mortality), only two cause-specific. Assessed for interactivity, Flesch readability, WCAG accessibility and transparency. Limitation: eight sites is a thin sample and search-engine screening is not reproducible. Relevance: moderate - public health-impact maps ignore sub-annual and sub-city variation, exactly what dense networks measure.

doi: 10.19191/ep26.4-5.s1.a1026.089

Atlante Aria e Salute Delphi co-creation 2026 Epidemiol Prev

Co-creation as a decision-making tool in the PNC 'Atlante Aria e Salute' project to define the contents of an integrated atlas

Two Delphi rounds (16 then 22 experts and regional stakeholders) plus plenaries to fix the Italian national air-and-health atlas: data sources (ISTAT cause-specific mortality, hospital discharges), territorial disaggregation, pollutant-outcome pairs graded on causal certainty, public-health relevance and routine-data accuracy, attributable-fraction methods and policy counterfactuals. Limitation: process paper; the abstract is truncated before the final outcome list, and no exposure model is specified. Relevance: low-moderate - a national open-data atlas is a natural consumer of fused monitor/LCS exposure surfaces if its resolution is fine enough.

doi: 10.19191/ep26.4-5.s1.a952.090

Bhattarai India crop burning inventories 2026 Atmos Environ

Air Quality Impacts of Post-monsoon Crop Residue Burning in Northern India Based on Regional and Global Emission Inventories

METADATA ONLY. No abstract at harvest. Title indicates a comparison of regional and global emission inventories for post-monsoon crop-residue burning and the resulting air-quality impact in northern India. Recorded for coverage. Relevance: moderate - inventory spread is the dominant uncertainty in Indo-Gangetic smoke attribution, and dense sensors are the observational constraint that could arbitrate between inventories.

doi: 10.1016/j.atmosenv.2026.122423

Canada wildfire smoke guidance 2026 Can J Public Health

Public health guidance about wildfire smoke: Four targets for improvement

Commentary proposing four distinct targets for improving Canadian wildfire-smoke guidance: a stronger evidence base (subpopulations, long-term effects, protective interventions), ethically deliberated recommendations, evaluation of guidance effectiveness beyond self-report, and a process-and-content standard for authoritative guidance. Limitation: no data. Relevance: moderate - the evaluation target presupposes measured indoor and personal exposure during smoke events, which is where low-cost sensors already dominate the evidence.

doi: 10.17269/s41997-026-01267-x

East Asia $PM_{2.5}$ O_3 warming levels 2026 Environ Pollut

Uneven Future $PM_{2.5}$ and O_3 responses to Emission Reductions under Different Warming Levels in East Asia

METADATA ONLY. No abstract at harvest. Title indicates climate-penalty-modulated $PM_{2.5}$ and O_3 responses to emission cuts in East Asia. Recorded for coverage. Relevance: low.

doi: 10.1016/j.envpol.2026.129354

GBD 2023 G20 stroke air pollution temperature 2026 Thromb Haemost

Estimates, temporal evolution, and contributors of the stroke burden due to air pollution and non-optimal temperature across G20 nations, 1990-2023

GBD 2023, G20 nations. Age-standardised stroke mortality attributable to air pollution plus non-optimal temperature fell from 52.81 to 17.46/100,000 (1990-2023; EAPC -3.44%, 95% CI -3.57 to -3.30); DALY rate EAPC -3.29%. The decline is driven by household air pollution; ambient PM and heat burdens rose or stalled, highest in low/middle-SDI members, and ARIMA projections to 2040 keep them flat. Limitation: GBD re-tabulation with joint attribution of two risk classes; the ARIMA projection adds no exposure information. Relevance: low - confirms that ambient PM, not household fuel, now carries the residual stroke burden.

doi: 10.1055/a-2977-5555

Global atmospheric Pb 2000-2019 2026 Environ Pollut

Global modelling of atmospheric lead (Pb) concentrations and health risks during 2000-2019

METADATA ONLY. No abstract at harvest. Title indicates global modelling of atmospheric Pb and associated risk over two decades. Recorded for coverage. Relevance: low.

doi: 10.1016/j.envpol.2026.129346

Global heatwave-fire smoke coexposure 2026 Sci Adv

Rising global population coexposure to heatwaves and landscape fire-sourced air pollution

Global 2000-2019: 55.9% of the population lived where heatwave-smoke coexposure increased; 916 million people/year had at least one coexposure day, rising by 287 million per decade; mean coexposure days per person rose from 1.0 to 1.7. Of the 5.5 billion person-day/decade increase, 70.5% was intensity and 29.5% population growth; low-HDI countries carried the most. Limitation: the fire-PM product is model-derived with no stated threshold sensitivity, and 'coexposure day' is a binary that discards concentration. Relevance: moderate - joint heat-smoke days are also when optical sensors are most biased (low RH, fresh smoke), so the exposure most in need of measurement is the hardest to measure cheaply.

doi: 10.1126/sciadv.aea9749

Gujral Dominici Memphis datacenter $PM_{2.5}$ 2026 Preprints.org (preprint)

Behind-the-Meter Versus Grid Electricity for AI Datacenters: Magnitude, Geography, and Equity of $PM_{2.5}$-Attributable Mortality in Memphis

Two datacenters 12 km apart. At a common 240 MW, grid-supplied C1 carried 10.83 attributable deaths/y (95% interval 7.83-13.84) vs 3.67 (2.65-4.68) for behind-the-meter C2; at real/permitted capacities the order reversed, 11.35 (6.48-17.29) for C2 vs 8.10 (4.60-12.38) for C1. 94% of C1 deaths fell >250 km away; 82% of C2 deaths within 50 km, concentrated in the low-income, high people-of-colour stratum. Limitation: preprint; permitted capacity is not operating capacity, and on-site turbine emission factors and the reduced-form dispersion model are the whole result. Relevance: high - a near-source burden that a local LCS network around the site could verify directly.

doi: 10.20944/preprints202610.0819.v1

Isfahan synbiotic omega-3 RCT 2026 J Res Med Sci

Effects of synbiotic versus omega-3 supplementation on total antioxidant capacity in adolescents exposed to ambient air pollution: A randomized double-blind controlled trial

70 healthy adolescents aged 10-18, 8 weeks synbiotic (n=38) vs omega-3 (n=32). TAC rose within both arms (839.8 to 903.4 and 898.6 to 953.3 umol/L) with no between-group difference (p=0.75). Limitation: active-comparator trial with no placebo arm, so within-arm rises cannot be separated from regression to the mean or seasonal change; exposure is not measured, so 'exposed to air pollution' is residence in Isfahan. Relevance: low.

doi: 10.4103/jrms.jrms_753_25

Landscape fire smoke health action plans review 2026 Public Health Chall

Health Management Planning for Landscape Fire Smoke Events: A Literature Review and Suggested Human-Centred Framework for Resource Development

Reviews smoke-event health management plans for at-risk individuals in Australia, Canada and the US and finds only four resources, two of them dedicated landscape-fire-smoke action plans. Proposes a design-thinking, human-centred framework for producing tailored plans for priority populations and clinicians. Limitation: a non-systematic search of grey literature; the framework is a process proposal with no evaluation of uptake or exposure reduction. Relevance: moderate - action plans hinge on local, timely $PM_{2.5}$ triggers, which is where corrected LCS (e.g. AirNow Fire and Smoke Map) carry operational weight.

doi: 10.1002/puh2.70428

Lu $PM_{2.5}$ longevity country panel specification analysis 2026 Front Public Health

Paradoxical $PM_{2.5}$-longevity estimates in global country panels are sensitive to differential trend control: a specification analysis, 2000-2023

4,406 country-years, two-way fixed effects with GDP, health spending and urbanisation. The conventional model gives a paradoxical +0.406 y life expectancy per 10 $\mu g\,m^{-3}$ $PM_{2.5}$ (0.144-0.668); economy-specific linear trends remove it (-0.023 y, -0.175 to 0.129), as do first differences (-0.006 y, -0.085 to 0.073). A future-exposure lead (+2 y) was positive alone (0.296, 0.042-0.549) but attenuated with current exposure. Limitation: the trend-adjusted estimates are null rather than negative, so the paper diagnoses bias without recovering the expected harm; national averaging discards almost all exposure contrast. Relevance: moderate - a methods warning for any ecological $PM_{2.5}$ panel, including those fed by sensor-network or satellite exposure surfaces.

doi: 10.3389/fpubh.2026.1936760

Shandong integrated air-pollutant GHG inventory 2026 Atmosphere

Strategies for Co-Control of Air Pollutants and Greenhouse Gases: Insights from an Integrated Emission Inventory in Shandong Province

2024 integrated inventory for 16 Shandong prefectures, 10 air pollutants and 4 GHGs over 8 source categories. Power/heat and industry gave 86.84% of CO_2; industry dominated SO_2, CO and VOCs; mobile sources NOx; agriculture NH_3, CH4 and N2O; fugitive dust dominated primary PM. Interspecies correlations show partial synergy, so co-control cannot be assumed from shared sources. Limitation: inventory only, no uncertainty bounds in the abstract, and fugitive-dust PM is the least constrained category in any inventory. Relevance: low-moderate - fugitive dust dominance is a coarse-mode signal that $PM_{10}/PM_{2.5}$ ratios from LCS can check locally.

doi: 10.3390/atmos17101016

Sharma France net-zero benefits 2026 Environ Int

Health and economic benefits of air quality improvements in France through diverse net-zero transition scenarios by 2050

METADATA ONLY. No abstract at harvest. Title indicates a health-impact and economic assessment of air-quality co-benefits under several French net-zero pathways to 2050. Recorded for coverage; no estimate is inferred. Relevance: low - scenario work, but it fixes the concentration ranges a future network would need to resolve.

doi: 10.1016/j.envint.2026.110555

Urban tree bark PM retention functional traits 2026 Environ Pollut

Species-specific PM retention by urban tree bark is associated with functional traits under controlled exposure

Modified open-top chambers compared PM capture by bark, leaves and branches of 12 calibration species. Per unit area bark retained more PM than foliage in 11 of 12 (Cinnamomum camphora 364.5 vs Juniperus formosana 112.1 ug/cm2). Upscaled to whole plants, bark $PM_{2.5}$-10 retention was 1.40x that of leaves while $PM_{2.5}$ was comparable - a coarse-selective filter. Lenticel density, roughness and groove prominence predicted interspecies variation; applied to 26 Hangzhou species the spread was 4.83-fold. Limitation: retention mass under controlled exposure says nothing about net ambient concentration reduction, and independent validation was limited. Relevance: low for sensing - deposition-mitigation claims of this type need ambient gradient measurement to be credible.

doi: 10.1016/j.envpol.2026.129262

Yu multicountry $PM_{2.5}$ admissions 2026 Lancet Planet Health

Cause-specific hospital admissions associated with short-term $PM_{2.5}$ exposure: a multicountry time-series analysis

346.5 million admissions from 949 communities in six countries/territories, 2000-2019; quasi-Poisson time series pooled by meta-analysis. Per 10 $\mu g\,m^{-3}$, all-cause admissions RR 1.080 (1.074-1.086), with effects persisting to 21 days. Malignant neoplasms (1.108, 1.094-1.123), mental disorders (1.100) and nervous system disease (1.085) all exceed respiratory (1.067, 1.060-1.074). Attributable fraction against the WHO daily 15 $\mu g\,m^{-3}$ fell from 8.9% (2000-09) to 3.9% (2010-19). Limitation: a 21-day distributed lag on cancer admissions invites harvesting and scheduling artefacts - an elective oncology admission is not an acute PM event - and heterogeneity across countries is described as substantial but the pooled RR is reported without it. Relevance: high - the long-lag, non-respiratory signal argues for continuous rather than episodic sensing.

doi: 10.1016/j.lanplh.2026.101526

Zhang US wildfire smoke five-year mortality impulse response 2026 medRxiv (preprint)

Wildfire smoke leaves a large five-year imprint on mortality in the United States

County-month mortality and wildfire-$PM_{2.5}$ panels 2011-2023; episodic smoke pulses are used to deconvolve overlapping responses and recover an impulse-response function. Excess mortality peaks about three years after a pulse and is gone by five. Integrated over 2011-2023 this gives 1.7 million smoke-associated deaths (95% UI 1.1-2.3 million), roughly ten times what long-term ambient $PM_{2.5}$ relative risks imply; circulatory and respiratory causes contribute comparably, adults >=65 dominate, and responses were larger during COVID-19. Limitation: a three-year lag in an ecological panel is exposed to slow confounders (migration, economic shocks, drought, heat) co-varying with fire years; the order-of-magnitude excess needs independent replication before it enters burden accounting. Relevance: high - a multi-year lag makes historical smoke-$PM_{2.5}$ reconstructions, including LCS-corrected ones, the binding input.

doi: 10.64898/2026.10.15.26365428

Other clinical endpoints — 7 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

China fire-sourced $PM_{2.5}$ cancer 2026 Environ Pollut

Extreme, Accumulated, and Lagged Exposure to Fire-Sourced $PM_{2.5}$ and Site-specific Cancer Risk in China

Fire-sourced $PM_{2.5}$ was positively associated with several site-specific cancers, with lagged and accumulated metrics showing the broadest pattern (thyroid, kidney, breast, lung, leukaemia, lymphoma) and extreme exposure the narrowest (strongest for larynx). Against all-source $PM_{2.5}$, fire $PM_{2.5}$ associations were stronger for thyroid, kidney, larynx, lymphoma and oral cancers, and stronger in the north and in low-GDP areas. Limitation: no effect sizes, n or geographic unit in the abstract; many sites x three metrics is a multiple-comparison surface, and thyroid cancer incidence is notoriously driven by screening intensity, which tracks region and income. Relevance: moderate - source-resolved smoke again outperforms mass.

doi: 10.1016/j.envpol.2026.129343

Dahl air pollution bone fractures high-risk population 2026 Environ Int

Air pollution and bone fractures in a high-risk population: differences by age and sex

METADATA ONLY. No abstract at harvest. The title indicates an analysis of ambient air pollution and fracture risk in a population at high fracture risk, stratified by age and sex. Recorded for coverage; no exposure model, design or estimate is inferred. Relevance: low-moderate - fracture is an under-studied PM endpoint, and bone-health associations have so far rested on residential annual means.

doi: 10.1016/j.envint.2026.110570

Ma green space pollution aging 2026 Environ Health Perspect

Associations of Green Space and Air Pollution Mixture with Heterogeneous Biological Aging Patterns

11,760 adults aged 20-65 with repeated measures, accelerated longitudinal design; three BAA trajectories. Pollution mixture OR 1.16 (1.08-1.24) for accelerated-aging membership, driven mainly by O_3 and PM_{10}; NDVI OR 0.84 (0.75-0.95), 25% of which was mediated through lower pollution. Limitation: trajectories differ in level, not slope, so the design measures cross-sectional BAA differences dressed as trajectories; exposure contrast within one city is narrow and residential NDVI is strongly SES-patterned. Relevance: moderate - the narrow within-city contrast is the case for intra-urban sensor density. Held from the 9 Oct issue.

doi: 10.1021/ehp.6c00004

Northern Thailand NPC survival $PM_{2.5}$ and coarse PM 2026 J Clin Med

The Effect of Fine ($PM_{2.5}$) and Coarse (PM_{10}-2.5) Particulate Matter on Survival Outcomes in Nasopharyngeal Carcinoma Patients: A Retrospective Cohort Study in a Northern Thailand Tertiary Care Center

METADATA ONLY. No abstract at harvest. Title indicates a single-centre retrospective cohort of nasopharyngeal carcinoma patients in northern Thailand relating fine and coarse PM to survival. Recorded for coverage; no estimate inferred. Relevance: moderate - northern Thailand's biomass-burning season is a well-instrumented low-cost-sensor region, and a coarse-fraction prognosis question needs PM_{10}-2.5 that optical LCS estimate poorly.

doi: 10.3390/jcm15197367

Ran $PM_{2.5}$ MASLD proteins 2026 J Adv Res

Association of protein signature of fine particulate matter with MASLD: the underlying pathways and Mendelian randomization

52,190 participants free of metabolic dysfunction-associated steatotic liver disease at baseline, 624 incident cases over a median 11.8 years. $PM_{2.5}$ gave HR 1.19 (1.10-1.30) and a 700-protein elastic-net signature of $PM_{2.5}$ gave HR 1.19 (1.08-1.31), the proteins enriched in NF-kB signalling and cytokine-receptor interaction; Mendelian randomisation supported CD28, CSF3 and HAVCR2 as causally associated with MASLD, and the signature mediated 15% (2.7-37.7%) of the $PM_{2.5}$-MASLD association. Limitation: a 700-protein signature trained on the same cohort in which it is then tested will absorb variance from everything $PM_{2.5}$ correlates with - smoking, adiposity, deprivation - so the mediated fraction is an upper bound, and its CI nearly spans an order of magnitude; the MR arm addresses protein-to-MASLD causality, not exposure-to-protein. Exposure is an annual mean from an interpolated surface, which is the weakest term in the chain. Relevance: moderate - liver endpoints are under-studied, and 15% mediation at best leaves the mechanism mostly open.

doi: 10.1016/j.jare.2026.10.021

SEER wildfire $PM_{2.5}$ head and neck cancer 2026 Laryngoscope

Wildfire Air Pollution Exposure and Head and Neck Cancer Incidence

SEER 2016-2022, county-level DLNM with 0-10-year lags, adjusted for tobacco, alcohol and demographics with county random effects. Cumulative RR per additional smoke-exceedance day 1.06 (1.04-1.09); any annual exposure 1.41 (1.24-1.60); per percentage-point of days >5 $\mu g\,m^{-3}$ 1.23 (1.13-1.33). Larynx largest (1.13, 1.07-1.19 per day), oropharynx 1.09. Limitation: ecological county-level exposure and covariates, a ten-year lag fitted on seven years of incidence, and wildfire-exposed western counties differ in screening and HPV epidemiology; the p16 stratification is the right control but its result is truncated in the abstract. Relevance: moderate-high - a second smoke-cancer signal this week (cf. 13 Oct China), both favouring upper aerodigestive sites.

doi: 10.1002/lary.70874

Zhang air pollution cataract meta-analysis 2026 Front Med (Lausanne)

Air pollution exposure and cataract outcomes: a systematic review and pollutant-specific meta-analysis

Fifteen observational studies, seven pooled with REML random effects (search to 10 Aug 2026; ROBINS-E). $PM_{2.5}$: HR 1.036 (0.988-1.086) per 1 $\mu g\,m^{-3}$, I2 99.7%; NO_2: HR 1.041 (1.013-1.070) per 10 $\mu g\,m^{-3}$, I2 86%; black carbon per IQR: HR 1.010 (0.983-1.038), I2 33%. Highest-vs-lowest quartile was positive for NO_2 (HR 1.094, 1.057-1.133) and null for $PM_{2.5}$ and PM_{10}; Knapp-Hartung widened the NO_2 interval. Limitation: with I2 near 100% the pooled $PM_{2.5}$ point estimate is not interpretable as a common effect - it averages incompatible exposure models and outcome definitions across too few studies. Relevance: low-moderate - the NO_2 > $PM_{2.5}$ ordering points at traffic proximity, which a dense network resolves better than satellite $PM_{2.5}$.

doi: 10.3389/fmed.2026.1969759

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
ALFA+ TRAP CSF GFAP	Air pollution is associated with sex-specific longitudinal changes in astrocytic and neuroinflammatory processes linked to cognitive decline in Alzheimer's disease at-risk individuals	*Alzheimers Dement*	Prospective cohort	LUR NO2, PM_{10}, $PM_{2.5}$, $PM_{2.5}$ absorbance	Spain (Barcelona, ALFA+)
Jin wildfire carbonaceous ADRD	Carbonaceous Components Underlying Wildfire Smoke $PM_{2.5}$-Associated Alzheimer's Disease and Related Dementias in Older US Adults	*medRxiv (preprint)*	Retrospective cohort	Annual smoke vs non-smoke $PM_{2.5}$ and smoke EC/OC (high-resolution model)	United States (Medicare, nationwide)
Khan Pakistan neurovascular review	Air Pollution and Neurovascular Health in Pakistan: An Emerging Public Health Crisis in South Asia - A Narrative Review	*Curr Neurovasc Res*	Narrative review	$PM_{2.5}$, PM_{10}, UFP (literature)	Pakistan
Shanghai prenatal UFP postpartum depression	Prenatal exposure to ultrafine particles and positive screening for postpartum depression: a prospective pregnancy cohort in Shanghai, China	*Environ Pollut*	Prospective cohort	Residential trimester-specific UFP number and $PM_{2.5}$ (modelled)	China (Shanghai)
Cardiovascular & metabolic					
Chen Lanzhou $PM_{2.5}$ components CVD	Acute cardiovascular mortality risks from $PM_{2.5}$ and its chemical components in an industrial valley city: evidence from Lanzhou, China	*Int J Environ Health Res*	Time-series (GAM) with mixture models	Daily $PM_{2.5}$; random-forest-predicted components (PAHs, metals, ions)	China (Lanzhou)
Huang gut microbiota mediation air pollution metabolic syndrome	The mediating roles of gut microbiota in the associations of air pollution and meteorological factors with metabolic syndrome	*iMetaOmics*	Microbiome mediation analysis	Long-term residential $PM_{2.5}$ and NO2; ambient temperature	China (cohort location not stated)
MESA incident AF time-varying pollution	Association of Air Pollution Exposure and Risk of Incident Atrial Fibrillation: The Multi-Ethnic Study of Atherosclerosis	*Heart Rhythm*	Prospective cohort	Spatiotemporal-model $PM_{2.5}$, NO2, NOx, O3 at residence; 2-y rolling means updated every 30 d	United States (6 MESA metros)
MESA long-term air pollution suPAR null	Long-term ambient air pollution is not associated with circulating soluble urokinase plasminogen activator receptor	*Environ Epidemiol*	Observational - cross-sectional	MESA Air spatiotemporal $PM_{2.5}$, NO2, NOx, O3, light-absorbing carbon (long-term residential)	United States (six MESA metros)
UKB arrhythmia NMR metabolome mediation	Large-scale plasma metabolic profiling identifies signatures and biomarkers associated with air pollution and arrhythmias: A prospective cohort study from UK Biobank	*Ecotoxicol Environ Saf*	Prospective cohort	LUR $PM_{2.5}$, PM_{10}, NO2, NOx at baseline residence	United Kingdom (UK Biobank)
UKB women deprivation $PM_{2.5}$-NO2 fatty liver	Socioeconomic disadvantage, air pollution, and metabolic health among women: analyses of UK Biobank and US data	*Front Endocrinol (Lausanne)*	Observational - cross-sectional	2010 residential $PM_{2.5}$-NO2 score (UK); state $PM_{2.5}$ (US)	United Kingdom (UK Biobank) / United States

First author	Focus	Journal	Design	Metric	Region
West Florida temperature x $PM_{2.5}$ CVD admissions	Joint Impacts of Ambient Air Pollution and Temperature on Cardiovascular Disease in West Florida	*J Am Heart Assoc*	Time-series	Daily network-monitor $PM_{2.5}$ and temperature, DLNM lags 0-7 with tensor-product interaction	United States (West Florida)
Reproductive & developmental					
Guangdong CHD $PM_{2.5}$ constituents	Differential Impacts of Fine Particulate Matter Constituents on Congenital Heart Disease across Critical Periconceptional Windows	*Environ Pollut*	Case-control study	Residential $PM_{2.5}$ mass and constituents (sulfate, nitrate, ammonium, BC, OM) by periconceptional window	China (Guangdong)
Hyperemesis gravidarum PM_{10} temperature	Seasonal ambient temperature and particulate matter as discriminators of hyperemesis gravidarum severity: a retrospective comparative observational study	*Front Public Health*	Observational - cross-sectional	Ambient PM_{10}, SO2 and temperature (regional)	Turkiye
Liang CHD circulating proteins	A nested case-control study of air pollution and congenital heart defects via circulating proteins	*Nat Commun*	Nested case-control	Periconception $PM_{2.5}$, PM_{10}, NO2; maternal and cord serum proteomics	China
Ni Wuhan birth cohort air pollution neurodevelopment target trial	Early Life Ambient Air Pollution, Sex Steroids and Neurodevelopment: Target Trial Emulation in a Prospective Cohort	*Environ Sci Technol*	Prospective birth cohort	Residential ambient pollutants (PM and gases) from preconception to age 2	China (Wuhan)
Pakistan cooking fuel undernutrition	Association between household fuel use and childhood undernutrition in Pakistan: evidence from a National survey	*Int J Environ Health Res*	Observational - cross-sectional	Household air pollution proxied by self-reported primary cooking fuel	Pakistan (national, PDHS 2017-18)
Placental angiogenic biomarkers review	Exposure to environmental toxicants during pregnancy and placental angiogenic biomarkers: An integrative review	*Reprod Toxicol*	Integrative review	$PM_{2.5}$, PM_{10}, NO2, O3 among other toxicants	Global / multi-region
Respiratory & allergic					
Guangdong COPD surveillance PM1 $PM_{2.5}$ PM_{10} PAF	Association between chronic respiratory diseases and the attributable risk of long-term exposure to atmospheric particulate matters among middle-aged and elderly Chinese population	*Fundamental Research*	Observational - cross-sectional	722-day mean PM1, $PM_{2.5}$, PM_{10} at 1 km (residential address, 2013-2019)	China (Guangdong)
Khan Gambia personal $PM_{2.5}$ pneumococcus	Longitudinal personal environmental exposure and Streptococcus pneumoniae colonisation in a cohort of children 5-14 years old in rural Gambia	*medRxiv (preprint)*	Prospective cohort	Personal 24-h $PM_{2.5}$ and CO monitors, repeated (8 visits)	Gambia (rural)
Kinzunga Kinshasa $PM_{2.5}$ respiratory	Acute Respiratory Impacts of Fine Particulate Matter ($PM_{2.5}$) in Kinshasa: Epidemiological Evidence and Pulmonological Implications	*Research Square (preprint)*	Time-stratified case-crossover	Daily $PM_{2.5}$ from PurpleAir and QuantAQ Modulair sensors plus bias-corrected GEOS-CF	Democratic Republic of the Congo (Kinshasa)
$PM_{2.5}$ constituents TB contacts	Differential effects of $PM_{2.5}$ constituents on tuberculosis infection and disease progression in 40,379 student close contacts	*Ann Am Thorac Soc*	Metadata only (no abstract)	$PM_{2.5}$ constituents	Not stated
Sittitoon Thailand haze $PM_{2.5}$ symptoms inequality	Health inequalities in $PM_{2.5}$-related symptoms during Thailand's haze season: the role of education, sociodemographic, lifestyle, protective behaviors, and environmental factors	*Front Public Health*	Cross-sectional survey	Provincial monthly $PM_{2.5}$ from satellite-based products	Thailand (nationwide)
Taiwan greenness pollution child asthma	Association of Urban Greenness and Air Pollution on Asthma Visits Among Children and Adolescents in Industrial and Non-Industrial Cities in Taiwan	*J Asthma Allergy*	Metadata only (no abstract)	Ambient air pollution (unspecified) and greenness	Taiwan
Trachalaki IPF $PM_{2.5}$	Are Short-Term Increases in $PM_{2.5}$ Linked to Respiratory Hospitalizations in IPF?	*Ann Am Thorac Soc*	Metadata only (no abstract)	Short-term $PM_{2.5}$	Not stated
Yoon pollutant asthma immune mechanisms review	Impacts and Immune Mechanisms of Pollutant-Mediated Asthma and Airway Inflammation	*Allergy Asthma Immunol Res*	Narrative review	PM, DEP, O3, NO2, SO2, VOCs, bioaerosols	Not stated

First author	Focus	Journal	Design	Metric	Region
Mechanistic toxicology					
Chen toluene soot toxicity	Oxidative toxicity and cytotoxicity of toluene combustion soot particles mediated by their microstructure	*Atmos Environ*	Metadata only (no abstract)	Laboratory toluene-flame soot	Laboratory
Comis Turin oxidative potential	Oxidative Potential and Elemental Fingerprints of Urban and Suburban Particulate Matter in Turin	*Atmosphere*	Chamber / laboratory	PM_{10} oxidative potential (ascorbic acid assay); total and bioaccessible metals	Italy (Turin)
EPFR size-resolved PM	Size-Dependent Characteristics, Exposure Risk, and Formation Mechanisms of Environmentally Persistent Free Radicals in Atmospheric Particulate Matter	*Environ Toxicol Chem*	Field measurement	EPFRs (EPR) in $PM_{2.5}$, PM_{10}, TSP; SiO2-catechol-ZnO simulations	China
Edwards macrophage superoxide SOA biomass plastic burning PM	Superoxide Release from Macrophages upon Exposure to Secondary Organic Aerosols, Biomass, and Plastic Burning Particulate Matter	*ACS Omega*	In vitro exposure	Laboratory SOA (limonene, toluene, naphthalene), biomass- and plastic-burning PM suspensions	Not applicable (laboratory)
Gut-liver axis pollutants MASLD review	The Role of the Gut-Liver Axis in the Development of Liver Diseases Induced by Selected Environmental Pollutants: A Narrative Review	*Int J Mol Sci*	Narrative review	PM among heavy metals and EDCs (toxicological literature)	Not applicable
Houston UFP PAH deposition	PAH-associated health risk from ultrafine particles: an urban field study	*Air Qual Atmos Health*	Field measurement	UFP respiratory deposition (MALDA) and size-segregated UFP-bound PAHs	United States (Greater Houston)
Inhaled polystyrene microplastics pulmonary fibrosis review	Polystyrene-associated pulmonary fibrosis: a summary based on the available evidence and prospects	*Front Immunol*	Narrative review	Airborne polystyrene micro/nanoplastics (inhalation)	Not applicable
Lung-on-a-chip pollutant toxicity review	Microfluidic lung-on-a-chip for evaluating inflammation and toxicity from environmental pollutants	*Front Bioeng Biotechnol*	Narrative review	Atmospheric PM, pathogens, VOCs, aromatic hydrocarbons, micro/nanomaterials (in vitro)	Laboratory
Mouth-throat replica deposition	The influence of model selection and flow rate on aerosol transport in replica models of the mouth and throat: An experimental study	*J Aerosol Sci*	Metadata only (no abstract)	Aerosol deposition in mouth-throat replicas	Laboratory
$PM_{2.5}$ HCC miR-25	$PM_{2.5}$ promotes migration of hepatocellular carcinoma by targeting miR-25	*Transl Cancer Res*	Animal + in vitro	$PM_{2.5}$ exposure of MHCC97H cells and mouse tumour models	Laboratory
$PM_{2.5}$ NLRP3 inflammasome lung cancer review	Possible Links Between $PM_{2.5}$-Induced NLRP3 Inflammasome Activation and Its Role in Lung Cancer Progression and Drug Resistance	*Int J Mol Sci*	Narrative review	$PM_{2.5}$ (toxicological literature)	Not applicable
$PM_{2.5}$ thyroid vitamin C rats	Vitamin C attenuates $PM_{2.5}$-induced thyroid dysfunction and mitophagy dysregulation by rebalancing ER-mitochondrial Ca2+ dynamics	*Ecotoxicol Environ Saf*	Animal + in vitro	$PM_{2.5}$ instillation (rats) and FRTL5 thyrocyte exposure	Laboratory
Time-restricted feeding $PM_{2.5}$ adipose metabolism	Metabolic Reprogramming by Time-Restricted Feeding Confers Resistance to $PM_{2.5}$-Induced Metabolic Disturbances in Adipose Tissue	*J Nutr Biochem*	Metadata only (no abstract)	$PM_{2.5}$ exposure (experimental; protocol not stated)	Not stated
Yoon PRL-3 ezrin airway epithelium particulate matter	PRL-3-sensitive cytoskeletal signaling mediates early airway epithelial responses to particulate matter	*Respir Res*	Animal + in vitro	Urban PM (UPM), diesel exhaust particles, PM_{10} reference materials	Not applicable (laboratory)
Yu alveolar organoid $PM_{2.5}$	Differential toxicity of water- and DMSO-extractable $PM_{2.5}$ fractions in human alveolar organoids	*Front Toxicol*	In vitro exposure	Water- vs DMSO-extractable $PM_{2.5}$ fractions	Laboratory (human alveolar organoids)
Occupational & indoor					
Creosote railroad switch PAH workers	Assessing PAH exposure during construction work of railroad switches using creosote-impregnated ties	*Ann Work Expo Health*	Field measurement	Full-shift breathing-zone PAH (16 EPA priority; particle + vapour), dermal tape strips, urinary 1-HP	Sweden

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Dental scaling bioaerosols	Divergent bacterial signatures in dental clinic bioaerosols: a real-field investigation	*J Hosp Infect*	Field measurement	Optical particle sizer; Andersen six-stage culturable and filter bioaerosol sampling	Not stated (dental clinic)
Global IAQ open standards framework	A unified framework for global indoor air quality: transitioning from guidelines to actionable open regulatory standards	*Build Environ*	Metadata only (no abstract)	Indoor air quality metrics (incl. PM)	Global / multi-region
Indoor NPF from outdoor O3	Elevated Outdoor O3 Can Sustain New Particle Formation and Growth Indoors	*Environ Sci Technol*	Field measurement	Indoor particle number size distributions; ventilated O3; limonene addition	Not stated (aged unoccupied office, high-O3 region)
LiFePO4 workers cardiac autonomic	Association between occupational LiFePO4 exposure and cardiac autonomic function: a cross-sectional study of cathode material production workers	*Occup Environ Med*	Metadata only (no abstract)	Occupational LiFePO4 particle exposure	Not stated
Office air purifier field CADR	Air purifier performance in an occupied office: Field CADR and effectiveness from a reconstructed device-off condition	*Build Environ*	Metadata only (no abstract)	Indoor PM with portable air cleaner (field CADR)	Not stated
Portable air purifiers airborne bacteria AMR offices schools	Response of culturable bacteria and their resistance to air purification across compartments	*Front Public Health*	Bioaerosol monitoring campaign	Culturable airborne, surface and human-associated bacteria before/after portable purifiers	Spain (Valencia) + Ireland (Dublin)
Simkovich LPG stove RCT quality of life	Cooking With Liquefied Petroleum Gas or Biomass and Quality of Life: Results From a Randomized Controlled Trial	*Indoor Air*	Randomised controlled trial	Personal $PM_{2.5}$ exposure (exposure-response analysis)	Not stated in abstract (biomass-cooking households)
Uberlandia street vendors $PM_{2.5}$ micronuclei panel	Urban air pollution and genotoxic damage in street vendors: a longitudinal repeated-measures panel study	*Int J Hyg Environ Health*	Panel study	PM1, $PM_{2.5}$, PM_{10} at seven fixed work-area monitoring points	Brazil (Uberlandia)
Walter neighbourhood air exchange inhaled dose children	Neighborhood-scale disparities in air exchange rates and inhaled pollution dose among school-aged children	*J Expo Sci Environ Epidemiol*	Metadata only (no abstract)	Residential air exchange rate and infiltrated PM / inhaled dose	United States
Sensing, forecasting & instrumentation					
Bachour WRF-Chem AOD post-correction hot desert	Statistical post-correction of WRF-Chem aerosol optical depth under hot-desert conditions	*Atmos Pollut Res*	Metadata only (no abstract)	WRF-Chem AOD vs ground photometry	Not stated
CONIOPOL CL61 aerosol typing	New operational perspective to identify aerosols and hydrometeors in real-time with a pioneering algorithm	*Atmos Meas Tech*	Instrument development	Volume depolarisation ratio and attenuated backscatter (Vaisala CL61 ceilometer)	Belgium (Uccle)
Chyzhykov gravimetric filter water Karl Fischer bias	Impact of Mass Loading and Water Speciation on Precision and Artifacts in Gravimetric Measurements of Atmospheric Particulate Matter	*Research Square (preprint)*	Laboratory metrology	Gravimetric PM1, $PM_{2.5}$, PM_{10} on 47 mm glass-fibre filters; Karl Fischer coulometric filter water	Poland (Krakow steelworks; Zdzieszowice coal)
Drone boundary-layer profiles Germany	Influence of irradiance and mixing layer height on the vertical trace matter distribution in the lower planetary boundary layer - drone-based investigation	*Atmos Chem Phys*	Field measurement	Drone vertical profiles of particle mass and number, CO2, O3 to 500 m	Germany (rural)
Dust lidar asymptotics	Asymptotic behavior of lidar scattering properties of absorbing dust aerosols across Rayleigh and geometrical-optics regimes: theory and implications	*Atmos Chem Phys*	Theoretical / optical modelling	Lidar ratio, depolarisation ratio, SSA vs size parameter and iron-oxide fraction	Not applicable (theory)
Epple DLR tire-wear particle chassis dynamometer	Temperature and Inflation Pressure Effects on Airborne Tire Wear Particle Measurements on a Chassis Dynamometer	*Atmosphere*	Laboratory measurement	Airborne TRWP number (MCPC 7 nm-2 um; OPS 0.3-10 um) and OPS mass proxy	Germany (DLR)

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Fluorescence-Raman-Mie lidar $PM_{2.5}$ profiles Beijing	Classification-constrained retrieval of $PM_{2.5}$ vertical profiles from fluorescence-Raman-Mie polarization lidar	*Atmos Meas Tech*	Remote sensing / lidar retrieval	Lidar extinction to $PM_{2.5}$, aerosol typed by depolarisation ratio and fluorescence capacity	China (Beijing)
HMAC message authentication vs LoRa telemetry poisoning	Securing AI-Driven Systems for Environmental Sustainability in IoT Applications: A Lightweight Message-Authentication Scheme Against Telemetry Poisoning	*Sensors (Basel)*	Method / algorithm	Network $PM_{2.5}$ telemetry (public datasets) used for forecast-poisoning test	Not applicable
He dust-storm PM_{10} deep learning	Transport-aware spatiotemporal deep learning for accurate PM_{10} forecasts during dust storms	*Atmos Environ*	Metadata only (no abstract)	PM_{10} during dust storms	Not stated
Kalisa East Africa passive sampler network PAHs metals	First Results from a Passive Air Monitoring Network across Five East African Countries for Polycyclic Aromatic Hydrocarbons (PAHs) and Metals	*Environ Pollut*	Passive-sampler network campaign	PUF passive samplers: 23 PAHs, 24 trace metals (total and water-soluble)	East Africa (five countries, 10 sites incl. Bukavu)
MISR Research Product Algorithm 1.1 km AOD	The MISR Research Product Algorithm - producing global, self-consistent, pixel-level aerosol retrievals for the Multi-angle Imaging SpectroRadiometer	*Atmos Meas Tech*	Satellite retrieval evaluation	MISR AOD, Angstrom exponent, particle shape and SSA at 1.1 km	Global
Nobin Bangladesh GNN $PM_{2.5}$	Explainable spatiotemporal Graph Neural Networks: Multi-City $PM_{2.5}$ forecasting in Bangladesh with geographic graph topology validation	*PLoS One*	Machine-learning model	Daily $PM_{2.5}$, 12-city network, 1-3 day forecasts	Bangladesh (12 cities)
Qian MAG-MISD probabilistic multi-station $PM_{2.5}$ forecasting	MAG-MISD: A Mode-Specific Adaptive Graph Framework for Multiscale Interaction and Station Dependence in Probabilistic $PM_{2.5}$ Forecasting	*Preprints.org (preprint)*	Deep-learning forecasting	Hourly station $PM_{2.5}$, 127 Yangtze River Delta monitors, 8 years	China (Yangtze River Delta)
SEARO air-quality early-warning review	Prediction accuracy of available early warning systems for air pollution across WHO South-East Asia region: a systematic review	*Front Public Health*	Systematic review	Forecast $PM_{2.5}$ and other pollutants (RMSE, R2, categorical accuracy, bias)	South-East Asia (India, Bangladesh, Thailand)
Saharan fresh dust optics	Optical properties and shape-dependent retrievals of the imaginary part of the refractive index of freshly emitted Saharan dust	*Atmos Chem Phys*	Field measurement	Size-resolved SSA, AAE, asymmetry, MAE/MSE; retrieved imaginary refractive index	Morocco (Sahara)
Schafer AERONET lunar AOD assessment	An assessment of lunar photometry in AERONET	*Atmos Meas Tech*	Instrument evaluation	Night-time lunar AOD (440-1640 nm), Angstrom exponent, SDA fine/coarse AOD	Global / multi-region (492 AERONET sites)
Shai Nanozen RT vs gravimetric	Comparison of respirable dust concentrations between gravimetric weighing and real-time dust monitoring equipment in non-mining industries	*Ann Work Expo Health*	Field measurement	Respirable dust: Nanozen DustCount 9000 optical vs its built-in filter cassette	South Africa (Gauteng)
Stutz Fairbanks inversion layers	Vertical distribution of pollution trapped by wintertime surface inversions in Fairbanks, Alaska, during the ALPACA experiment	*Atmos Chem Phys*	Field measurement	Vertical trace-gas profiles (LP-DOAS) with 1-D chemistry-transport model	United States (Fairbanks, Alaska)
Sylvester PurpleAir Philadelphia	Validation of PurpleAir $PM_{2.5}$ Sensors in Temperate Urban Environments	*ACS EST Air*	Field measurement	PurpleAir PA-II $PM_{2.5}$ (cf_1 vs atm outputs) co-located with reference; RH from PA and weather station	United States (Philadelphia)
Taiwan transboundary PM_{10} forecast	Forecasting transboundary PM_{10} pollution in northern Taiwan using upwind observations and transport timing	*Environ Pollut*	Machine-learning model	PM_{10} at 3 receptors; 9 continental upwind records; trajectory timing composite	Taiwan (northern)
UAV rotor-wash bias in airborne pollutant sensing	UAV-Induced Measurement Bias in Air Pollution Sensing: A Physics-Informed Framework for Characterization, Sensor Placement, and Correction	*Sensors (Basel)*	UAV instrument deployment	UAV-mounted $PM_{2.5}$, PM_{10} and gas sensors vs fixed reference	Not stated
Wildfire smoke onset detection fixed cameras	Detecting Wildfire Smoke Onset from Fixed View Camera Networks	*Sensors (Basel)*	Method / algorithm	Visible smoke plumes in fixed-camera video (no mass metric)	United States (California) + transfer sets

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Exposure assessment & modelling					
Chen Chengdu WSON partitioning	Simultaneous measurements of gas- and aerosol-phase water-soluble organic nitrogen in the winter urban atmosphere of Chengdu, China	*Atmos Chem Phys*	Field measurement	Gas- and particle-phase WSON and amines; aerosol liquid water content	China (Chengdu)
Cheng BTH multi-source satellite $PM_{2.5}$ stacking	Machine Learning-Based Retrieval of $PM_{2.5}$ Concentrations Using Multi-Source Satellite Data in the Beijing-Tianjin-Hebei Region of China	*Atmosphere*	Machine-learning model	Satellite AOD + LST, NDVI, tropospheric NO2 + meteorology -> surface $PM_{2.5}$	China (Beijing-Tianjin-Hebei)
China 20 cities building green $PM_{2.5}$	City-size-dependent interactive effects of building and green-space on $PM_{2.5}$ pollution: Evidence from 20 representative Chinese cities	*Environ Pollut*	Metadata only (no abstract)	$PM_{2.5}$ vs urban morphology	China (20 cities)
DC 1 km WRF-SMOKE-CMAQ	Neighborhood-Scale O3, NO2, and $PM_{2.5}$ in Washington, D.C. from a 1 x 1 km2 WRF-SMOKE-CMAQ Framework	*ACS EST Air*	Chemical transport model	Hourly O3, NO2, $PM_{2.5}$ at 1 km (2017)	United States (Washington, D.C.)
Delhi-NCR COVID transitions	Time series evidence on air quality transitions under urban shock during COVID-19 shutdown in Delhi-NCR, India	*Sci Prog*	Monitoring-record analysis	Daily $PM_{2.5}$, PM_{10}, NOx, SO2, CO, O3 at 10 CAAQMS stations	India (Delhi-NCR)
Five megacities $PM_{2.5}$ O3 trends meteorology 2018-2023	Urban Air Quality Trends and Meteorological Influences Across Five Global Megacities, 2018-2023	*Atmosphere*	Monitoring-record analysis	Monitor $PM_{2.5}$, PM_{10}, O3, NO2, SO2 with temperature, SLP, wind	Global (Shanghai, Delhi, Paris, Los Angeles, Sao Paulo)
Ghazaryan explainable ML annual $PM_{2.5}$ Europe Central Asia	Explainable Machine Learning for $PM_{2.5}$ Prediction and Comparative Country-Level Risk Assessment in Europe and Central Asia	*Preprints.org (preprint)*	Machine-learning model	Annual city $PM_{2.5}$ vs climatic and socio-economic indicators	Europe and Central Asia (10 cities)
Gray-green infrastructure $PM_{2.5}$ MSPA	Spatially Explicit Morphological Regulation of Gray-Green Infrastructure for $PM_{2.5}$ Mitigation in High-Density Cities: An Interpretable MSPA-Machine Learning Framework	*Build Environ*	Metadata only (no abstract)	$PM_{2.5}$ vs green-space morphology (MSPA)	Not stated
Hanoi $PM_{2.5}$-bound PCDD/Fs	Levels and contributing sources to $PM_{2.5}$-bound PCDD/Fs and dl-PCBs in ambient air of Hanoi	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$-bound PCDD/Fs and dioxin-like PCBs	Vietnam (Hanoi)
Harbin ammonium $PM_{2.5}$ and precipitation sources	From atmosphere to river: Tracing the origins and deposition of ammonium in both $PM_{2.5}$ and precipitation in Harbin	*Environ Pollut*	Metadata only (no abstract)	$PM_{2.5}$ ammonium (isotopic source tracing) and wet deposition	China (Harbin)
Li Guangzhou 3D vegetation	Scale-Dependent Effects of Three-Dimensional Urban Vegetation on Land Surface Temperature and Air Pollutants: Synergy and Trade-Off Zones in Guangzhou, China	*Atmosphere*	Statistical model	1 km gridded $PM_{2.5}$, O3, NO2; layered vegetation volume	China (Guangzhou)
Lucknow Holika Dahan $PM_{2.5}$	Impact of Holika Dahan on ambient $PM_{2.5}$ and associated health risks in an urban environment of Lucknow, India	*Environ Monit Assess*	Field measurement	Night-time $PM_{2.5}$ mass and chemistry (tracers, ATR-FTIR), HYSPLIT, AERMOD	India (Lucknow)
Monsoon aerosol Pb isotopes rainwater offsets	Monsoon-related variability of water-soluble aerosol Pb isotopes and systematic aerosol-rainwater offsets at Peng Chia Yu	*Atmos Environ*	Metadata only (no abstract)	Water-soluble aerosol Pb isotopes vs rainwater	Taiwan (Peng Chia Yu)
Multi-scale $PM_{2.5}$ nonlinear time-series two metros	Deciphering multi-scale $PM_{2.5}$ temporal dynamics and cross-scale interactions in two climatically distinct metropolitan areas: insights from advanced nonlinear time-series analysis	*Atmos Pollut Res*	Metadata only (no abstract)	Monitor $PM_{2.5}$ time series (multi-scale decomposition)	Not stated
North Macedonia biomass PMF	Biomass burning drives urban $PM_{2.5}$ across the Western Balkans: evidence from a six-city source apportionment study in North Macedonia	*Environ Pollut*	Source apportionment	$PM_{2.5}$ from 1,195 daily filters, EPA PMF 5.0	North Macedonia (six cities)
OA molecular composition predicts ROS yield MCR-VK	Linking molecular composition of organic aerosols to reactive oxygen species formation using an MCR-VK framework	*Atmos Chem Phys*	Laboratory + field measurement	Filter $PM_{2.5}$: H2O2 + radical ROS yields; FT-ICR/Orbitrap OA composition	Multi-site (China, Germany, Finland) + chamber SOA
Pourkiaei Santiago CMAQ EDGAR vs INEMA inventories	High-Resolution Air Quality Modeling over Santiago, Chile: A Comparative Assessment of EDGAR and INEMA Anthropogenic Emission Inventories	*Research Square (preprint)*	Chemical transport model evaluation	CMAQ $PM_{2.5}$ (9/3/1 km nests), source apportionment and DDM sensitivities	Chile (Santiago)

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Pu Lima hybrid transfer-learning $PM_{2.5}$	A Hybrid Transfer Learning Approach for Ambient $PM_{2.5}$ Exposure Modeling in LMIC Contexts: A Case Study from Lima, Peru	*ACS EST Air*	Machine-learning model	Daily 1 km $PM_{2.5}$ from reference + low-cost monitors, MAIAC AOD, WRF-Chem and covariates	Peru (Lima)
Salem Riyadh $PM_{2.5}$ vegetation	Spatiotemporal variation of $PM_{2.5}$ and its relationship with vegetation cover and meteorological factors during spring (2016-2022) in Riyadh, Saudi Arabia	*Environ Monit Assess*	Monitoring-record analysis	Spring $PM_{2.5}$ from 16 regulatory stations; fractional vegetation cover	Saudi Arabia (Riyadh)
Sao Paulo 2024 fire extremes	Compound fire-meteorology extremes drove megacity air pollution during the 2024 South American fire season	*J Environ Manage*	Chemical transport model	Surface $PM_{2.5}$ and CO, AOT, trajectories; fire-on/fire-off global simulations	Brazil (Sao Paulo State)
Temirzhanova Pavlodar snow aerosol	Geochemical Characteristics of the Pollution of Snow Cover with Aerosols in an Industrial Center	*Atmosphere*	Field measurement	Elemental composition of snow-cover solid fraction (ICP-MS)	Kazakhstan (Pavlodar)
Torii Kalimantan peat fire super-linear scaling	Peat Burning Scales Super-linearly with Fire-Season Magnitude in the West Kalimantan-Sarawak Border Region, 2012-2025	*Research Square (preprint)*	Satellite time-series analysis	VIIRS active-fire detections on peat vs non-peat; FRP; back-trajectories to Sarawak receptors	Indonesia / Malaysia (Kalimantan-Sarawak)
US school wildfire smoke trends	Spatio-Temporal Trends in Wildfire Smoke Exposures at Schools and Childcare Centers in the United States and Implications for Child and Adolescent Health	*Environ Res*	Metadata only (no abstract)	Wildfire smoke $PM_{2.5}$ at school and childcare locations	United States
Wang China $PM_{2.5}$ decade	From rapid decline to slower improvement: Decadal nonlinear evolution and regional predictive associations of $PM_{2.5}$ in China	*Atmos Pollut Res*	Metadata only (no abstract)	Ambient $PM_{2.5}$ trends, 2015-2024	China (nationwide)
Zhang Fusion-DeepRSM satellite $PM_{2.5}$	Fusion-DeepRSM for Satellite-Derived $PM_{2.5}$ Estimation and Precursor Sensitivity Prediction	*Environ Sci Technol*	Machine-learning model	Gridded $PM_{2.5}$ from CTM + multisource satellite + ground fusion	China (nationwide)
Zhou condensation sink efficiency hygroscopicity	Condensation Sink Efficiency of Atmospheric Particles Regulated by Their Chemical Compositions and Hygroscopicity	*Environ Sci Technol*	Laboratory + field measurement	Pre-existing particle condensation sink toward H2SO4/HIO3; composition and hygroscopicity	Laboratory
Burden, policy & mitigation					
Abu Dhabi PM burden 2008-2022	Ambient Air Pollution Health Impacts in High-Income Countries, Part II: A Case Study in the United Arab Emirates	*Risk Anal*	Health impact model	Anthropogenic PM and ozone, 2019-2022 vs 2008	United Arab Emirates (Abu Dhabi)
Air-health web atlases review Italy	PNC project 'Atlante Aria e Salute': exploratory web review and quality assessment of open-access atlases on the health effects of air pollution	*Epidemiol Prev*	Web review	$PM_{2.5}$/PM_{10} (all platforms), NO2 and O3 (fewer)	Global / multi-region
Atlante Aria e Salute Delphi co-creation	Co-creation as a decision-making tool in the PNC 'Atlante Aria e Salute' project to define the contents of an integrated atlas	*Epidemiol Prev*	Delphi consensus	Outdoor $PM_{2.5}$, PM_{10}, NO2, O3 (atlas exposure layers)	Italy
Bhattarai India crop burning inventories	Air Quality Impacts of Post-monsoon Crop Residue Burning in Northern India Based on Regional and Global Emission Inventories	*Atmos Environ*	Metadata only (no abstract)	Modelled $PM_{2.5}$ from crop-residue burning	India (northern)
Canada wildfire smoke guidance	Public health guidance about wildfire smoke: Four targets for improvement	*Can J Public Health*	Commentary	Wildfire smoke (guidance)	Canada
East Asia $PM_{2.5}$ O3 warming levels	Uneven Future $PM_{2.5}$ and O3 responses to Emission Reductions under Different Warming Levels in East Asia	*Environ Pollut*	Metadata only (no abstract)	Projected $PM_{2.5}$ and O3 under warming levels	East Asia
GBD 2023 G20 stroke air pollution temperature	Estimates, temporal evolution, and contributors of the stroke burden due to air pollution and non-optimal temperature across G20 nations, 1990-2023	*Thromb Haemost*	GBD secondary analysis	GBD ambient PM and household solid-fuel pollution; non-optimal temperature	Global / multi-region (G20)
Global atmospheric Pb 2000-2019	Global modelling of atmospheric lead (Pb) concentrations and health risks during 2000-2019	*Environ Pollut*	Metadata only (no abstract)	Modelled airborne (particle-bound) Pb	Global / multi-region

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Global heatwave-fire smoke coexposure	Rising global population coexposure to heatwaves and landscape fire-sourced air pollution	*Sci Adv*	Exposure model (secondary data)	Landscape fire-sourced $PM_{2.5}$ (modelled) coincident with heatwave days	Global / multi-region
Gujral Dominici Memphis datacenter $PM_{2.5}$	Behind-the-Meter Versus Grid Electricity for AI Datacenters: Magnitude, Geography, and Equity of $PM_{2.5}$-Attributable Mortality in Memphis	*Preprints.org (preprint)*	Health impact model	Modelled population-weighted $PM_{2.5}$ from on-site turbines vs grid generation	United States (Memphis)
Isfahan synbiotic omega-3 RCT	Effects of synbiotic versus omega-3 supplementation on total antioxidant capacity in adolescents exposed to ambient air pollution: A randomized double-blind controlled trial	*J Res Med Sci*	Randomised controlled trial	Ambient air pollution (not measured at individual level)	Iran (Isfahan)
Landscape fire smoke health action plans review	Health Management Planning for Landscape Fire Smoke Events: A Literature Review and Suggested Human-Centred Framework for Resource Development	*Public Health Chall*	Literature review	Landscape fire smoke (no metric)	Australia, Canada, United States
Lu $PM_{2.5}$ longevity country panel specification analysis	Paradoxical $PM_{2.5}$-longevity estimates in global country panels are sensitive to differential trend control: a specification analysis, 2000-2023	*Front Public Health*	Ecological panel, fixed effects	Population-weighted annual ambient $PM_{2.5}$ (World Bank WDI)	Global / multi-region (186 economies)
Shandong integrated air-pollutant GHG inventory	Strategies for Co-Control of Air Pollutants and Greenhouse Gases: Insights from an Integrated Emission Inventory in Shandong Province	*Atmosphere*	Emission inventory	Primary PM, SO2, NOx, VOCs, NH3, CO plus CO2, CH4, N2O by source and city	China (Shandong)
Sharma France net-zero benefits	Health and economic benefits of air quality improvements in France through diverse net-zero transition scenarios by 2050	*Environ Int*	Metadata only (no abstract)	Scenario-modelled $PM_{2.5}$ and co-pollutants to 2050	France
Urban tree bark PM retention functional traits	Species-specific PM retention by urban tree bark is associated with functional traits under controlled exposure	*Environ Pollut*	Chamber / laboratory	Gravimetric PM retained on bark, leaves, branches ($PM_{2.5}$, $PM_{2.5}$-10)	China (Hangzhou)
Yu multicountry $PM_{2.5}$ admissions	Cause-specific hospital admissions associated with short-term $PM_{2.5}$ exposure: a multicountry time-series analysis	*Lancet Planet Health*	Multi-location time-series	Daily ambient $PM_{2.5}$ (community-level), lags to 21 days	Multi-country (Australia, Brazil, Canada, Chile, New Zealand, Taiwan)
Zhang US wildfire smoke five-year mortality impulse response	Wildfire smoke leaves a large five-year imprint on mortality in the United States	*medRxiv (preprint)*	Distributed-lag panel deconvolution	Monthly county-level wildfire-attributable $PM_{2.5}$ (smoke $PM_{2.5}$ surface)	United States (all counties)
Other clinical endpoints					
China fire-sourced $PM_{2.5}$ cancer	Extreme, Accumulated, and Lagged Exposure to Fire-Sourced $PM_{2.5}$ and Site-specific Cancer Risk in China	*Environ Pollut*	Ecological / registry analysis	Fire-sourced vs all-source $PM_{2.5}$; lagged, accumulated and extreme metrics	China (nationwide)
Dahl air pollution bone fractures high-risk population	Air pollution and bone fractures in a high-risk population: differences by age and sex	*Environ Int*	Metadata only (no abstract)	Ambient air pollution (metric not stated in record)	Not stated in record
Ma green space pollution aging	Associations of Green Space and Air Pollution Mixture with Heterogeneous Biological Aging Patterns	*Environ Health Perspect*	Prospective cohort	$PM_{2.5}$, PM_{10}, SO2, O3 mixture (WQS); 500 m NDVI	China (single high-pollution city)
Northern Thailand NPC survival $PM_{2.5}$ and coarse PM	The Effect of Fine ($PM_{2.5}$) and Coarse (PM_{10}-2.5) Particulate Matter on Survival Outcomes in Nasopharyngeal Carcinoma Patients: A Retrospective Cohort Study in a Northern Thailand Tertiary Care Center	*J Clin Med*	Metadata only (no abstract)	$PM_{2.5}$ and PM_{10}-2.5 (exposure model not stated)	Thailand (northern)
Ran $PM_{2.5}$ MASLD proteins	Association of protein signature of fine particulate matter with MASLD: the underlying pathways and Mendelian randomization	*J Adv Res*	Prospective cohort	Annual average $PM_{2.5}$ by bilinear interpolation, time-weighted; 700-protein elastic-net signature	Not stated (large prospective cohort, 52,190 participants)

First author	Focus	Journal	Design	Metric	Region
SEER wildfire $PM_{2.5}$ head and neck cancer	Wildfire Air Pollution Exposure and Head and Neck Cancer Incidence	*Laryngoscope*	Ecological / registry analysis	County-level wildfire-specific $PM_{2.5}$ (exceedance days, any exposure, % days > 5 μg/m^3), 0-10-y lags	United States (SEER counties)
Zhang air pollution cataract meta-analysis	Air pollution exposure and cataract outcomes: a systematic review and pollutant-specific meta-analysis	*Front Med (Lausanne)*	Systematic review & meta-analysis	Long-term $PM_{2.5}$, PM_{10}, NO2, black carbon (study-specific models)	Global / multi-region (15 studies)

Provenance and method

Period **11 – 17 October 2026**, Sunday to Saturday, **seven daily issues**, **123 records**, **123 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. Tier mix **15 A, 43 B, 65 C**. The 17 October daily and this rollup were built on the **19 October run** (the Saturday 23:00 run did not fire); no record is held. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.